

**INDUSTRY 4.0**  
**SMART GARMENT FACTORY**  
**SAFETY & MONITORING SYSTEM**  
[TaxyShield]

**Higher National Diploma in Software Engineering**

**Final Project Documentation**

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**National Institute of Business Management**  
**Kandy**

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**Authors:**

|                      |                 |
|----------------------|-----------------|
| W.A.S. Soysa.        | KAHNDSE252F-006 |
| T.A. A. Kaumadee.    | KAHNDSE252F-012 |
| W.A.D.N.S. Weragoda. | KAHNDSE252F-018 |
| T.D. Jayasooriya.    | KAHNDSE252F-020 |

**Name of the program:** Higher National Diploma in Software Engineering 25.2F

**Name of supervisor:** Mrs. N.W.A.M.V.A.D.T.Weerasinghe

**Name of Institution:** National Institute of Business Management

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“The project is submitted in partial fulfilment of the requirements of the Higher National Diploma of Software Engineering of National Institute of Business Management.”

## Declaration

“I declare that this project is original and does not include, without proper acknowledgement, any material previously submitted for a Higher National Diploma or similar qualification at any institution. To the best of my knowledge and belief, it contains no material previously published or written by another individual or myself, except where appropriate references are provided within the text. I also grant permission for this project report, if approved, to be made available for photocopying, for the title and abstract to be shared with external organizations.”

|                      |                 |  |
|----------------------|-----------------|--|
| W.A.S. Soysa.        | KAHNDSE252F-006 |  |
| T.A. A. Kaumadee.    | KAHNDSE252F-012 |  |
| W.A.D.N.S. Weragoda. | KAHNDSE252F-018 |  |
| T.D. Jayasooriya.    | KAHNDSE252F-020 |  |

Project Supervisor: Mrs. N.W.A.M.V.A.D.T.Weerasinghe.

Signature: .....

Date of Submission: .....

Course Director: Mrs. N.W.A.M.V.A.D.T.Weerasinghe.

Signature: .....

Date: .....

## Preamble

This project report documents the design, development, and implementation of an Industry 4.0 Smart Garment Factory Safety and Monitoring System. It details the integration of IoT hardware, AI-driven computer vision, and mobile application dashboards to ensure occupational safety and automated environmental control. The report follows a standard academic structure, moving from the organizational context and problem definition through methodology, analysis, design, and conclusion.

This expanded edition builds on the original submission by elaborating the reasoning behind each design decision, adding fuller technical explanations of the hardware-software interaction, and clarifying how each chapter connects to the overall goal of a proactive, largely autonomous industrial safety net. Where the original document used a placeholder or a compact bullet, this version explains the underlying engineering rationale so the report can also serve as a stand-alone technical reference.

## **Abstract**

Industrial environments face significant challenges regarding worker safety, environmental monitoring, and access control. This project presents a Smart Garment Factory Safety & Monitoring System that leverages the Internet of Things (IoT), Artificial Intelligence, and a mobile application to automate safety protocols. The system utilises an ESP32 microcontroller network with a PIR sensor, a Light Dependent Resistor (LDR), a temperature sensor, a gas sensor, and a current sensor to continuously monitor factory floor conditions. Six relay channels convert these readings into automated actions: lighting control, air-conditioning activation, gas-exhaust ventilation, a combined fire response (buzzer, water pump, AC/light shutdown), and an overcurrent power cut-off. A dedicated ESP32-CAM module scans worker QR codes to log precise In and Out timestamps to a secure digital ledger. An Android mobile application, built with Kotlin and Jetpack Compose and backed by Firebase Realtime Database, gives administrators and workers real-time visibility into sensor status, alerts, and attendance. Together, these components form a proactive, largely autonomous safety net for industrial floors.

## **List of Keywords**

- Industry 4.0
- Internet of Things (IoT)
- Smart Garment Factory
- Worker Safety
- ESP32
- ESP32-CAM
- Artificial Intelligence
- Android
- Firebase
- Sensors
- Relay Automation
- QR Attendance

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## List of Acronyms and Abbreviations

| Acronym | Meaning                           |
|---------|-----------------------------------|
| IoT     | Internet of Things                |
| PIR     | Passive Infrared (motion sensor)  |
| LDR     | Light Dependent Resistor          |
| AC      | Air Conditioner                   |
| ppm     | Parts Per Million                 |
| UID     | Unique Identifier                 |
| UML     | Unified Modelling Language        |
| UI      | User Interface                    |
| QR      | Quick Response (code)             |
| API     | Application Programming Interface |

## **Acknowledgement**

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We would also like to express our sincere appreciation to Marsylka Manufacturing Lanka (Pvt) Ltd for approving us to conduct this academic project within their organization. In particular, we are grateful to their Human Resources Manager for reviewing and approving our request, and for permitting us to observe relevant workplace activities and collect the information necessary for this project. Their cooperation and trust in providing access to the factory environment were invaluable in grounding this project in a real industrial setting.

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## Chapter 1: Introduction

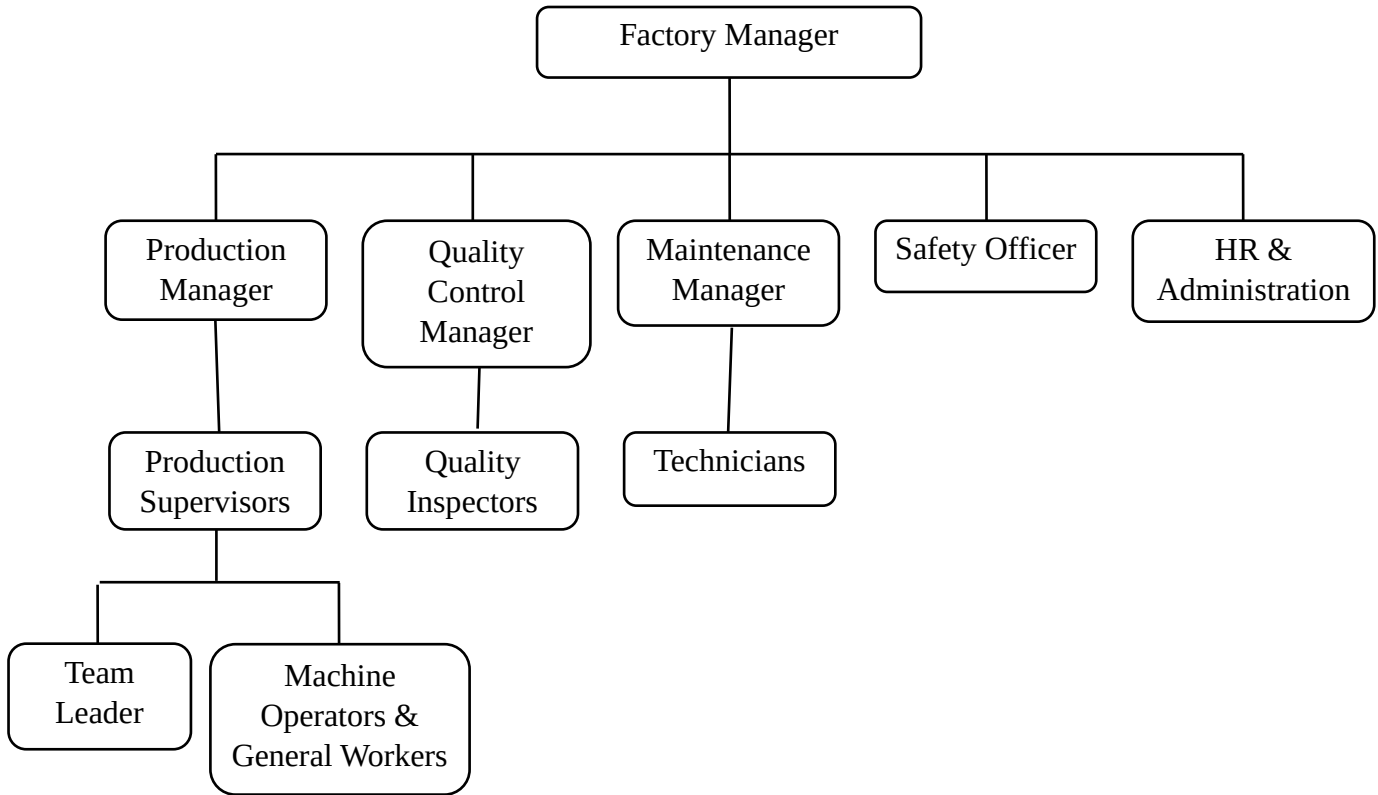
### 1.1 Introduction of the Organization

Marsylka Manufacturing Lanka (Pvt) Ltd is a garment manufacturing company located in the Export Processing Zone, Polgahawela, Sri Lanka, specialising in the production of ladies' wear. The factory operates with a workforce of approximately 750 employees across 14 active production lines, making it a mid-to-large-scale apparel manufacturing operation with significant exposure to the everyday safety and machine-reliability challenges typical of the garment industry. As a supplier operating under international buyer compliance requirements, the factory maintains formal safety policies and procedures, but as detailed in the sections below, several of its safety and monitoring processes remain manual, reactive, or fragmented across standalone devices rather than unified into a single digital system. This project uses Marsylka Manufacturing Lanka as its industry case study, with the factory's Quality Assurance representative providing the operational data that grounds the requirements and design decisions throughout this report.



## 1.2 Organization Structure

The following diagram shows the organization's structure.



*Figure 1.1: Organisation structure of Marsylka Manufacturing Lanka (Pvt) Ltd*

### **1.3 Current Operations in Organization**

Machine maintenance at the factory is currently reactive, with servicing performed "only when a fault occurs" rather than on a fixed weekly, monthly, or scheduled basis. Common problems reported include thread breakage, needle breakage, fabric jamming, and motor failure. Significantly, the factory's QA representative confirmed that these machine breakdowns directly affect daily production output. Fault reporting itself is entirely manual; a worker who notices a problem informs their supervisor verbally, with no maintenance team routing, log book, or digital system currently in place to track recurring issues or response times.

On the safety side, the factory already has a workplace safety policy, conducts regular safety training, and requires workers to wear PPE including face masks and gloves. Emergencies such as fire, smoke, or gas leaks are handled by following established fire procedures aligned with buyer compliance requirements, maintaining designated smoke areas, and providing fire-safety training to employees. For monitoring, the factory currently uses CCTV cameras, smoke sensors, temperature sensors, and a fire alarm system, but these operate as separate, disconnected devices rather than as part of an integrated digital safety monitoring system, and the factory does not currently use gas sensors or motion sensors at all. This combination of reactive machine maintenance and fragmented, non-integrated safety devices is the core operational gap this project addresses.

## 1.4 Users and Responsibilities of the Organization

- **Quality Assurance / Admin:** Oversees compliance with safety and quality standards across all 14 production lines, and under the proposed system would monitor real-time dashboards, machine-health data, and receive critical alerts for fire, gas, or equipment hazards.
- **Floor Supervisor:** Currently the first point of contact when a worker reports a machine fault or safety concern; would use the proposed mobile application to receive and act on real-time alerts for their assigned line.
- **Machine Operator / General Worker:** Operates sewing machines (lockstitch, overlock, flatlock, buttonhole, button-sewing, bartack, and zigzag machines) and is currently responsible for verbally reporting faults or hazards to a supervisor; relies on PPE, safety training, and the factory's fire procedures for protection while on the floor.

The QA representative specifically noted the difficulty of "handling 100% compliance procedures while achieving maximum efficiency in production" without it affecting safety, directly highlighting that current responsibilities are stretched thin across manual processes, which the proposed system aims to relieve through automation.

## 1.5 Problem Definition

Based on the factory's current operations, this project addresses the following problems:

- **Reactive, Undocumented Machine Maintenance:** With approximately 550 of the factory's sewing machines in active operation and servicing performed only when a fault occurs, recurring issues like thread breakage, needle breakage, and motor failure are not tracked systematically, and breakdowns are confirmed to directly disrupt daily production.
- **Fragmented Safety Technology:** The factory already operates CCTV cameras, smoke sensors, temperature sensors, and a fire alarm system, but these exist as standalone tools rather than a unified digital safety monitoring system. There is no gas sensing, no motion sensing, and no central dashboard tying these signals together.
- **Manual, Verbal Fault and Hazard Reporting:** Faults and safety concerns are reported by informing a supervisor directly, with no maintenance team routing, log book, or digital system, creating delays and no historical record for analysis.
- **Compliance vs Efficiency Pressure:** The QA representative identified fire risk and health risk as the most common safety concerns, and explicitly noted that balancing full safety compliance against production efficiency is an ongoing operational challenge.

## 1.6 Project Objectives

- To develop a digital safety monitoring system capable of consolidating the factory's existing sensor inputs (CCTV, smoke, temperature, fire alarm) with new gas and motion sensing into a single, real-time platform.
- To introduce automated, threshold-based alerts for fire, smoke, gas, and temperature hazards that reduce reliance on manual observation and verbal fault reporting.
- To provide a structured digital channel for logging and tracking machine faults (thread breakage, needle breakage, fabric jamming, motor failure), replacing the current supervisor-only verbal reporting process.
- To develop a mobile application for supervisors and management, a feature the factory's QA representative directly confirmed would be useful, giving real-time visibility into safety status and machine health across all 14 production lines.
- To support the factory's existing PPE, safety-training, and fire-compliance procedures with real-time monitoring, rather than replacing them, so the system strengthens rather than disrupts current buyer-compliance practices.

## **1.7 Proposed Solution**

The proposed solution is an AI-based Smart Garment Factory Safety Monitoring and Employee Assistance System, directly informed by the factory's confirmed interest in such a system, real-time safety alerts, and a supervisor/manager mobile application. Building on the sensors already present on the floor (CCTV, smoke, temperature, fire alarm), the system introduces gas and motion sensing to close the current monitoring gaps, and ties all sensor inputs into a unified dashboard rather than the disconnected devices currently in use. Threshold-based logic mirroring the escalation pattern the factory already follows for fire procedures triggers automated alerts and, where applicable, safety responses, while a digital fault-logging channel replaces the current verbal, supervisor-only reporting process for machine issues such as thread breakage, needle breakage, and motor failure. A companion mobile application delivers this information to supervisors and management in real time, directly addressing the compliance-versus-efficiency tension the QA representative raised, by giving staff visibility into safety conditions without requiring constant manual floor inspection.

## **1.8 Chapter Summary**

This chapter introduced Marsylka Manufacturing Lanka (Pvt) as the industry case study for this project, and outlined its current operations: reactive machine maintenance, manual verbal fault reporting, and safety technology (CCTV, smoke, temperature, and fire alarm systems) that operates without central integration or gas/motion sensing. These gaps, confirmed directly by the factory's Quality Assurance representative, define the problems this project sets out to solve, and directly shape the objectives and proposed AI-based safety monitoring and employee assistance system detailed in this report.

## Chapter 2: Methodology

### 2.1 Introduction

This chapter outlines the methodology adopted to design and develop the Smart Garment Factory Safety Monitoring and Employee Assistance System. It explains how requirements were gathered directly from Marsylka Manufacturing Lanka (Pvt) Ltd, the tools selected to build the hardware and software components, and the testing strategies used to validate the system before deployment. Given that the project addresses real safety and machine maintenance gaps identified at an operating garment factory, the methodology places particular emphasis on grounding every design decision in verified, factory-reported data rather than on assumptions.



## 2.2 Data Collection Methods

- **Structured Questionnaire (Primary Data Collection):** A structured Google Form questionnaire was distributed to Marsylka Manufacturing Lanka's Quality Assurance representative, covering respondent and factory information, machine information, worker safety practices, existing technology usage, and project feedback. This single, targeted response from a Dedicated Quality Assurance officer with direct floor-level visibility provided authoritative, first-hand data on the factory's 750 employees, 14 production lines, and approximately 550 active sewing machines, and formed the primary evidence base for the problem definition and objectives in Chapter 1.
- **Categorical and Open-Ended Questions:** The questionnaire combined closed multiple-choice questions (e.g., machine types in use, PPE requirements, existing safety technologies) with open-ended questions (e.g., "What are the most common safety risks in your factory?", "What challenges do you face in monitoring worker safety?"). This mixed format allowed quantifiable comparison of current practices against Industry 4.0 best practices, while still capturing qualitative insight into operational pain points such as the tension between compliance and production efficiency.
- **Permission and Ethics:** The questionnaire explicitly requested and received formal permission to use Marsylka Manufacturing Lanka as the case study for this HND final year project, and separately requested and received permission to photograph the factory environment and equipment for academic purposes only, ensuring the data collection process was conducted ethically and with documented organizational consent.
- **Software Process Model:** The Waterfall model was adopted for development. Hardware safety logic sensor thresholds, relay behaviour, and alert conditions were defined, built, and tested first, based directly on the risks the factory identified (fire risk and health risk) and the technologies it confirmed were already in place (CCTV, smoke sensors, temperature sensors, fire alarm system). The mobile application and reporting layer were developed only once this safety-critical foundation was stable, since the factory's own QA representative confirmed that machine breakdowns already disrupt daily production, meaning any new system had to be verified thoroughly before being trusted on a live floor.

## **2.3 Software Process Model**

The Waterfall model was adopted for the development of the TaxyShield system. This model was chosen because the hardware safety logic, sensor thresholds, and alert conditions needed to be strictly defined, built, and tested first. These foundational elements were based directly on the risks identified by the factory (fire risk and health risk) and existing technologies. Since the factory's QA representative confirmed that machine breakdowns disrupt daily production, the mobile application and reporting layer were developed sequentially only after the safety-critical hardware foundation was verified and stable.

## 2.4 Software Development Tools

The following software development tools and frameworks were used to implement the system:

1. Integrated Development Environments (IDEs)
  - Android Studio: Used for developing the Android mobile application.
  - Arduino IDE / PlatformIO: Used for writing and uploading the C++ firmware to the ESP32 and ESP32-CAM microcontrollers.
2. Mobile Application Framework
  - Kotlin & Jetpack Compose: The modern programming language and UI toolkit used to build the native Android application interfaces.
3. Database & Backend
  - Firebase Realtime Database: A cloud-hosted NoSQL JSON database used to sync live IoT sensor data and alerts in real-time.
  - Firebase Authentication: Used to securely manage worker and admin logins.
4. Artificial Intelligence & Machine Learning Tools
  - TensorFlow Lite / ML Kit: Used to run the pre-trained object detection models directly on the mobile device for PPE (mask) detection and AR features.
  - Google Colab: A cloud-based Jupyter notebook environment used for data pre-processing, image dataset handling, and training the Artificial Intelligence models (such as PPE mask detection) utilising Python and cloud GPUs.
  - Python / OpenCV: Used during the data pre-processing and AI training phase for health and risk detection.
5. UI/UX Prototyping
  - Figma: Used to design the wireframes and user interfaces for the Admin and Worker dashboards.

### 2.4.1 Emergent Technologies Used

To resolve the factory's challenges with reactive machine maintenance and fragmented safety protocols, this project integrates four pillars of Industry 4.0 technology into a unified platform designed to create a zero-accident, high-efficiency manufacturing environment:

#### 2.4.1.1 Artificial Intelligence & Machine Learning (AI/ML)

The system relies on a Python FastAPI backend running advanced deep learning models to automate safety monitoring and replace manual floor checks. Utilising the YOLOv8 and YOLO11 architectures for real-time computer vision, the system actively detects whether workers are wearing mandatory PPE (face masks and gloves) at factory entrances, tracks skeletal keypoints for instant fall detection, and analyses facial micro-expressions to monitor worker fatigue. Furthermore, to address the factory's reliance on reactive maintenance, an Ensemble Learning approach using a Random Forest Classifier (Scikit-Learn) analyses machine telemetry. This allows the system to predict the failure probability of sewing machines, warning mechanics of overheating or motor issues before a critical breakdown halts production.

#### 2.4.1.2 Internet of Things (IoT)

To digitise the physical factory floor, the system deploys a network of ESP32 microcontrollers equipped with environmental sensors (DHT22 for temperature, MQ2 for gas, PIR for motion) and hardware sensors (ACS712 for current). These nodes stream live telemetry to a Firebase Realtime Database, unifying previously disconnected CCTV and fire alarm systems into a single digital dashboard. Beyond passive monitoring, the IoT network is designed for automated response; the microcontrollers can physically trigger 6-channel relays to automatically activate exhaust fans, sound buzzers, or cut main power. This ensures that life-safety actions such as shutting off a sewing machine during an electrical overcurrent event happen instantaneously without waiting for human intervention.

#### 2.4.1.3 Augmented Reality (AR)

The Kotlin-based Android mobile application features an Augmented Reality (AR) Assistant module designed to drastically reduce machine downtime and simplify troubleshooting. When a mechanic points their smartphone camera at any of the 550 sewing machines on the factory floor, a CNN image classifier identifies the specific machine type. The application then bridges the gap between digital telemetry and the physical world by rendering holographic overlays directly onto the machine. Mechanics can view real-time sensor data, maintenance status, and digital repair instructions contextually and hands-free, eliminating the need to consult paper manuals or navigate complex menus while holding tools.

#### 2.4.1.4 Blockchain Technology

As a garment manufacturer subject to strict international buyer compliance requirements, maintaining a transparent and mathematically secure audit trail is critical. To achieve this, high-security events are processed by the backend and mined into a custom blockchain ledger. This includes Worker Attendance logs, captured via ESP32-CAM QR scanners, and the Safety Ledger, which records both worker-submitted Emergency SOS reports and automated AI hazard detections. By utilising blockchain technology, the system ensures that safety incidents, emergency response times, and attendance records remain immutable; they cannot be altered, deleted, or tampered with retroactively by either management or workers.

## 2.5 Testing Strategies

Testing is an essential part of the development process to ensure the reliability, quality, and performance of the system. Several testing strategies were applied to verify that all system components function correctly and meet user expectations.

### 2.5.1 Functionality Testing

Each sensor and its corresponding automated response were tested individually to confirm correct triggering behaviour; for example, verifying that a simulated temperature rise or gas presence correctly activates the intended alert, mirroring the fire-risk and health-risk concerns Marsylka's QA representative flagged as the factory's most common safety issues. Testing also covered the digital fault-logging function intended to replace the current practice of workers verbally informing a supervisor when a machine issue (thread breakage, needle breakage, fabric jamming, or motor failure) occurs.

### 2.5.2 Usability Testing

The mobile application's interface was evaluated for clarity, particularly from the perspective of supervisors and management, the two user groups the factory specifically confirmed would benefit from such an application. Feedback focused on ensuring safety alerts are immediately visible and unambiguous, and that the interface does not add friction to floor operations, addressing the QA representative's concern that maintaining full compliance can come at the cost of production efficiency.

### 2.5.3 Interface Testing

Communication between the sensor hardware, backend service, and mobile application was tested end-to-end to confirm that a triggered alert reaches the supervisor/manager application without noticeable delay, directly targeting the factory's existing gap of having safety devices (CCTV, smoke sensors, temperature sensors, fire alarm) that operate independently rather than as an integrated monitoring system.

#### 2.5.4 Compatibility Testing

The mobile application was tested across multiple Android OS versions and screen sizes to ensure supervisors and managers can reliably access safety information and alerts regardless of their device, supporting practical day-to-day use across a 750-employee, 14-line production facility.

## 2.6 Implementation Plan

- **Phase 1:** Requirements confirmation and threshold calibration, based directly on the factory's questionnaire responses (fire risk, health risk, and the machine faults most commonly reported: thread breakage, needle breakage, fabric jamming, motor failure).
- **Phase 2:** Sensor and hardware assembly, extending the factory's existing CCTV, smoke, temperature, and fire-alarm infrastructure with new gas and motion sensing.
- **Phase 3:** Firmware development for automated, threshold-based alert and relay logic.
- **Phase 4:** Development of the digital fault-logging and alert pipeline, replacing the current supervisor-only verbal reporting process.
- **Phase 5:** Mobile application development (Kotlin, Jetpack Compose) for supervisors and managers, integrated with the real-time database.
- **Phase 6:** End-to-end integration testing and validation against the specific risks and gaps identified by Marsylka Manufacturing Lanka, followed by deployment planning.



## **2.7 Chapter Summary**

This chapter described a Waterfall-based methodology in which every stage was informed directly by data collected from Marsylka Manufacturing Lanka (Pvt) Ltd, a 750-employee, 14-production-line ladies' wear manufacturer. Requirements were gathered through a structured questionnaire completed by the factory's Quality Assurance representative, with formal permission obtained to use the factory as the project's case study. Appropriate hardware and software tools were selected to extend the factory's existing but disconnected safety technologies into a unified, real-time monitoring system, and a staged testing strategy was defined for functionality, usability, interface, and compatibility to ensure the resulting system reliably addresses the fire risk, health risk, and machine-fault reporting gaps the factory itself identified.

## Chapter 3: Analysis

### 3.1 Introduction

This chapter analyses the requirements of the Smart Garment Factory Safety Monitoring and Employee Assistance System and visualises the interaction between the system's actors, Worker, Supervisor/QA Admin, Maintenance Technician, and the IoT Sensor System itself using UML diagrams. The analysis is grounded directly in the operational data gathered from Marsylka Manufacturing Lanka (Pvt) Ltd, ensuring the modelled use cases, data flows, and system behaviour reflect the factory's actual reporting practices, existing technology, and identified risks (fire risk and health risk).

## 3.2 UML Diagrams

### 3.2.1 Use Case Diagram of Current System

Before TexyShield, safety monitoring at the factory was entirely manual. Workers marked attendance on a paper sheet, supervisors relied on visual patrols to check safety conditions, and faults were reported verbally or by radio and logged by hand. The use case diagram below captures this as-is process, which the proposed system in the following sections is designed to replace.

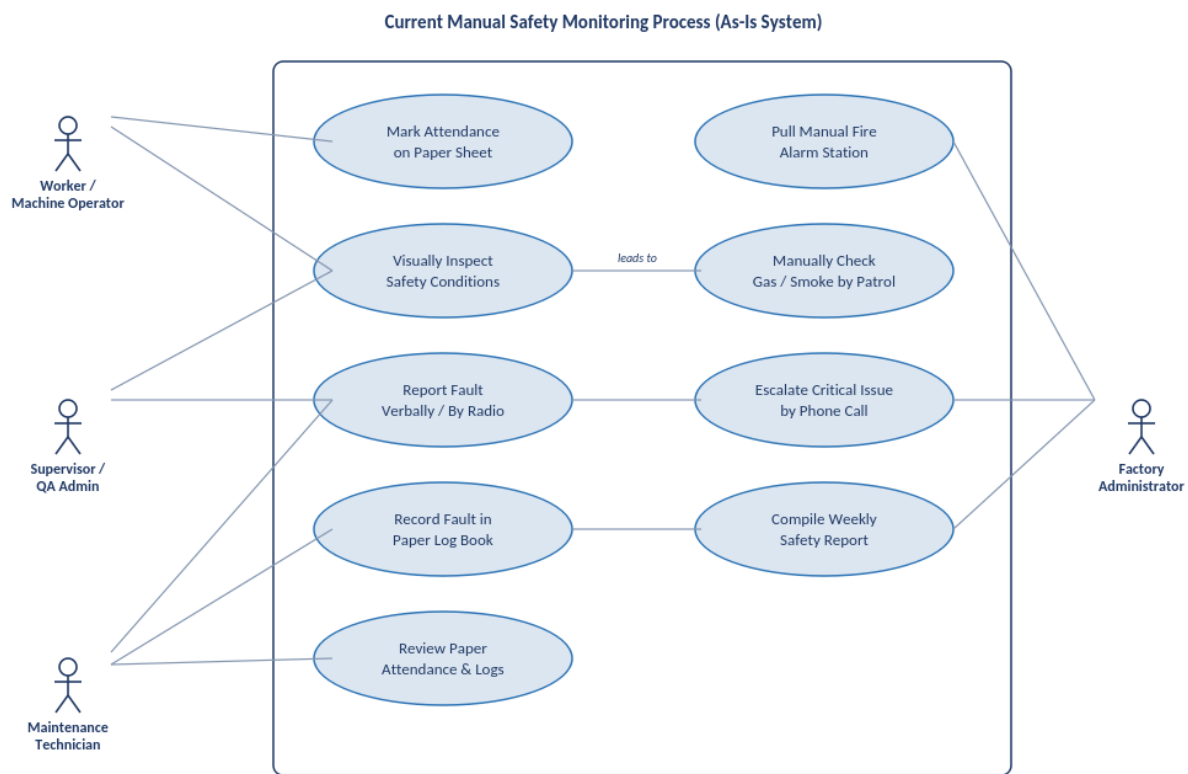


Figure 3.1: Use case diagram of the current (manual) system

### 3.2.2 Use Case Diagram of Proposed System

Shows four actors: Worker, Supervisor/QA Admin, Maintenance Technician, and the IoT Sensor System interacting with core functions such as QR attendance, fault reporting, live dashboard monitoring, and automated safety response. The IoT Sensor System acts autonomously, triggering alerts without waiting for human input.

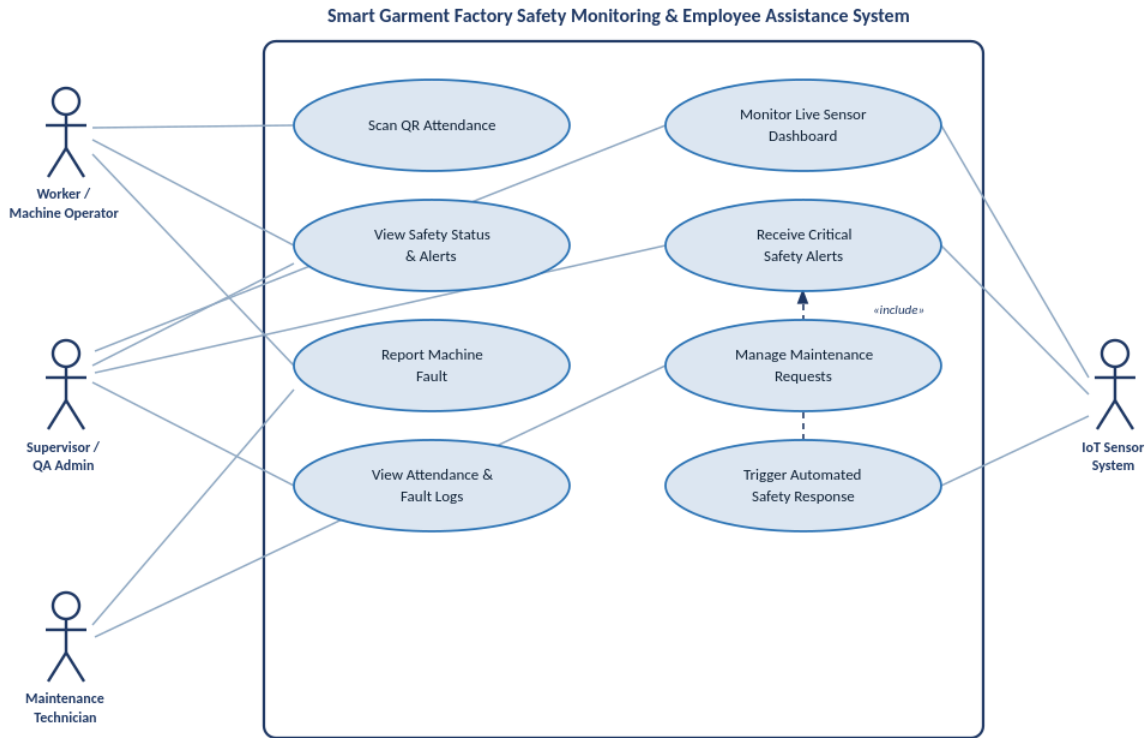


Figure 3.2: Use case diagram of system actors

### 3.2.3 Class Diagram of Proposed System

The class diagram below models the core data and behaviour of TaxyShield. A base User class is specialised into Worker, Supervisor and MaintenanceTechnician, each interacting with domain classes such as AttendanceRecord, MachineFault, SensorReading, SafetyAlert, MaintenanceRequest, BlockchainLedgerEntry, Actuator and EmergencyReport.

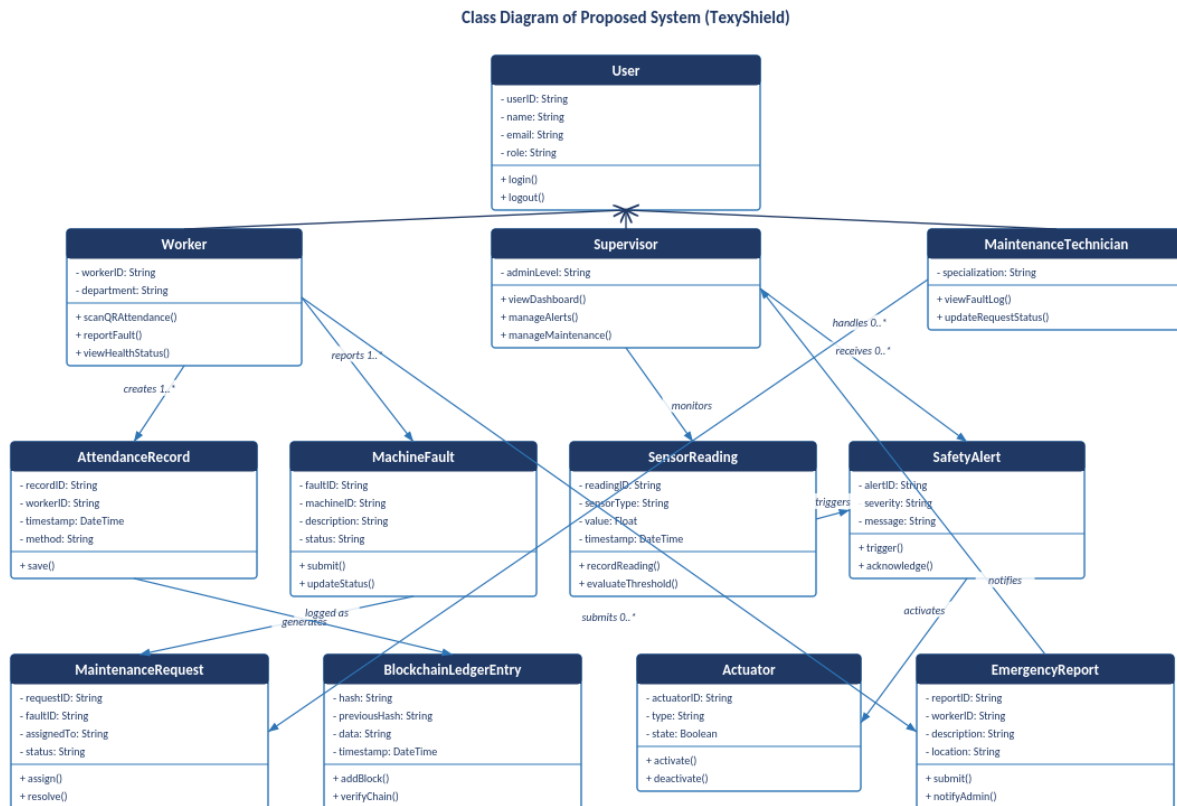


Figure 3.3: Class diagram of proposed system

### 3.2.4 Sequence Diagram of Proposed System [For each Use Case]

Traces a fire event from sensor trigger to resolution: temperature and gas readings are evaluated together, actuators respond instantly on-device, and the alert is only pushed to the Supervisor App afterwards so the physical safety response never depends on network availability.

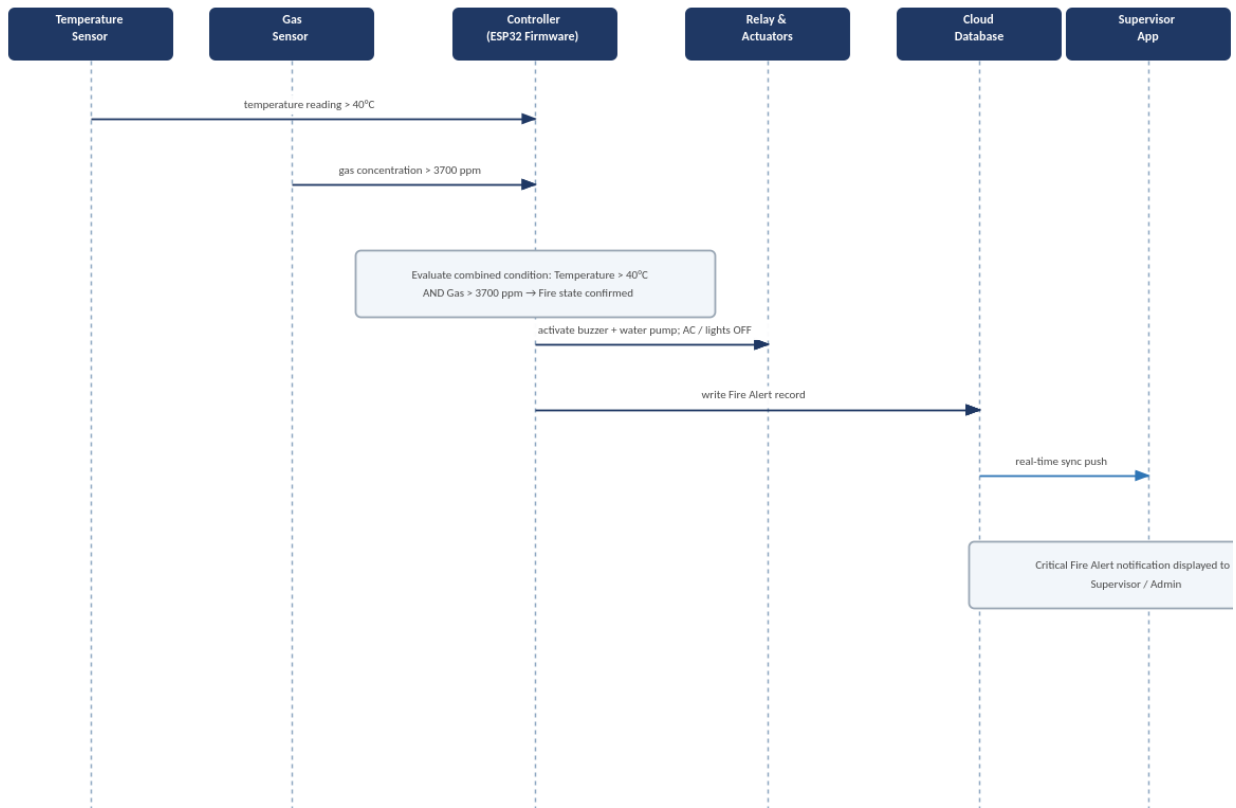


Figure 3.4: Fire detection sequence diagram

The sequence diagrams below trace the message flow for each use case identified in the proposed system's use case diagram (Figure 3.2), complementing the fire detection scenario already covered in section 3.2.3.

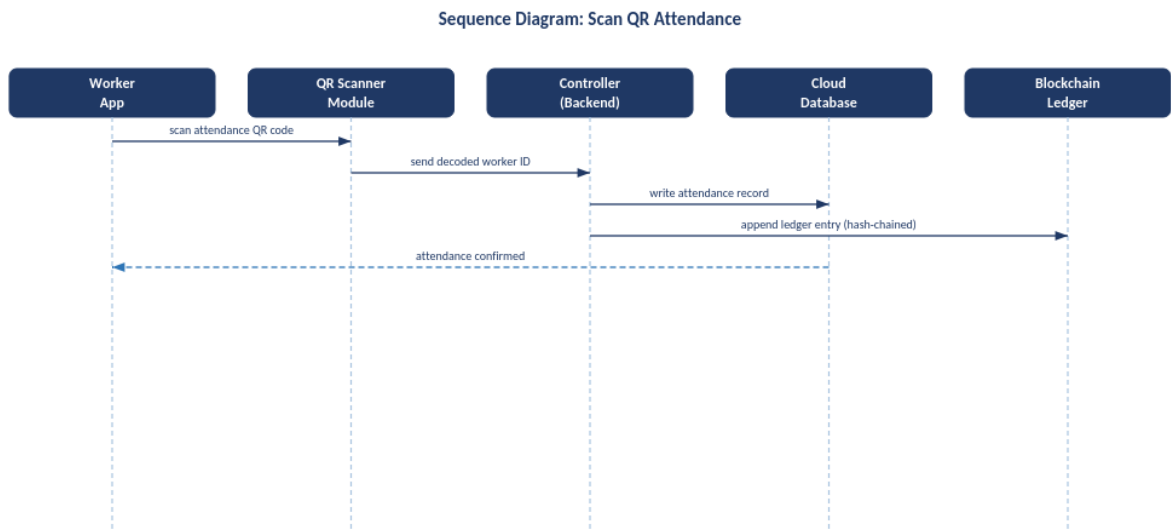


Figure 3.5: Sequence diagram: Scan QR Attendance

A worker scans the attendance QR code; the controller resolves the worker ID, writes the attendance record to the cloud database, and appends a hash-chained entry to the blockchain ledger.

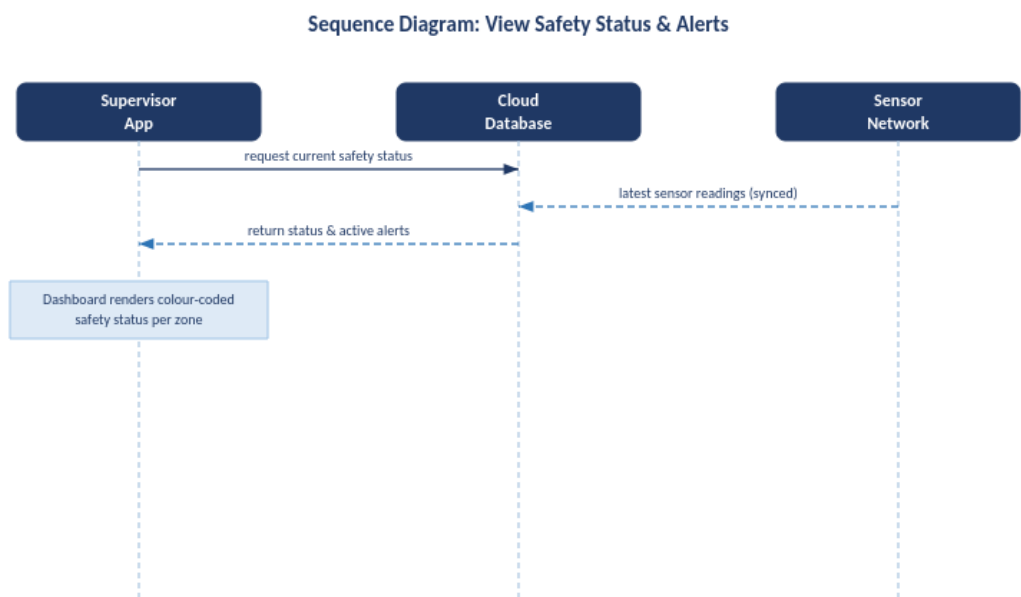


Figure 3.6: Sequence diagram: View Safety Status & Alerts

The Supervisor App requests the current safety status; the cloud database returns the latest synced sensor readings together with any active alerts for display on the dashboard.

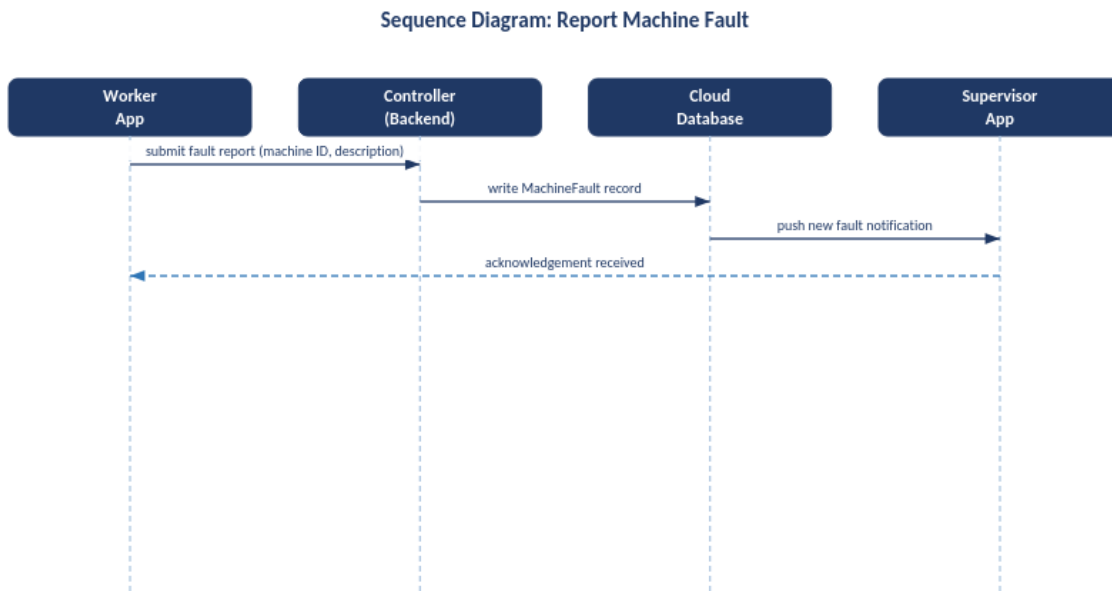


Figure 3.7: Sequence diagram: Report Machine Fault

A worker submits a fault report from the Worker App; the controller writes the fault record and the cloud database pushes a notification to the Supervisor App.

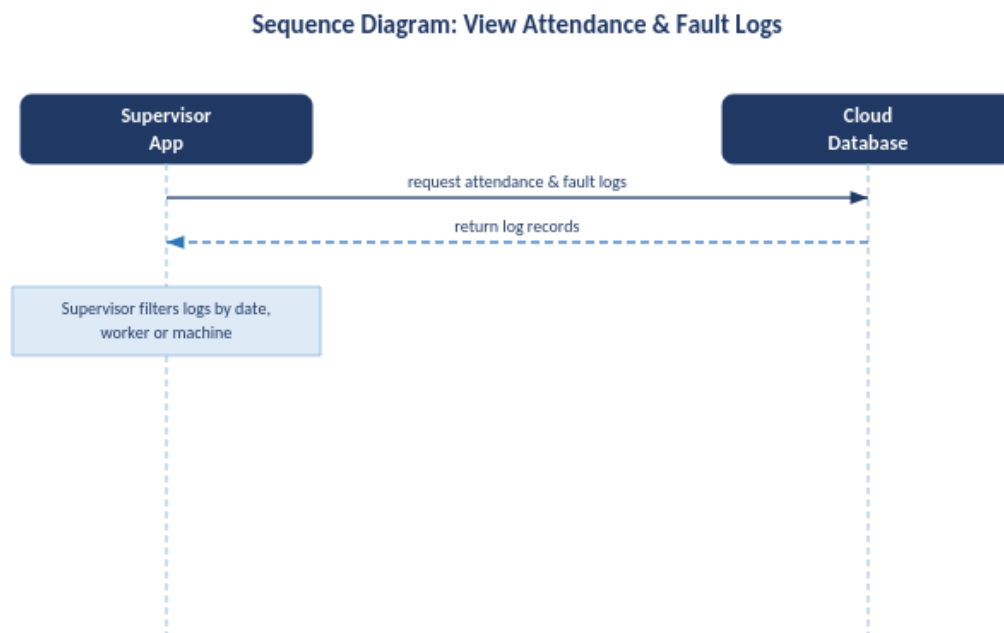


Figure 3.8: Sequence diagram: View Attendance & Fault Logs



The Supervisor App requests attendance and fault log records from the cloud database for filtering and review by date, worker or machine.

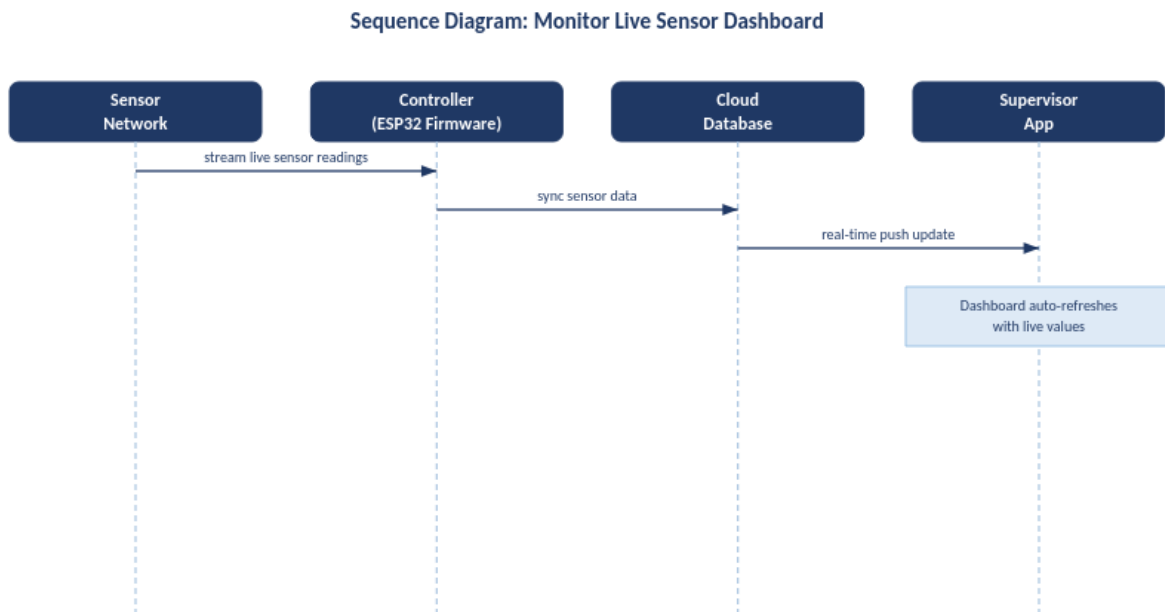


Figure 3.9: Sequence diagram: Monitor Live Sensor Dashboard

Live sensor readings stream from the sensor network through the controller to the cloud database, which pushes real-time updates to the Supervisor App dashboard.

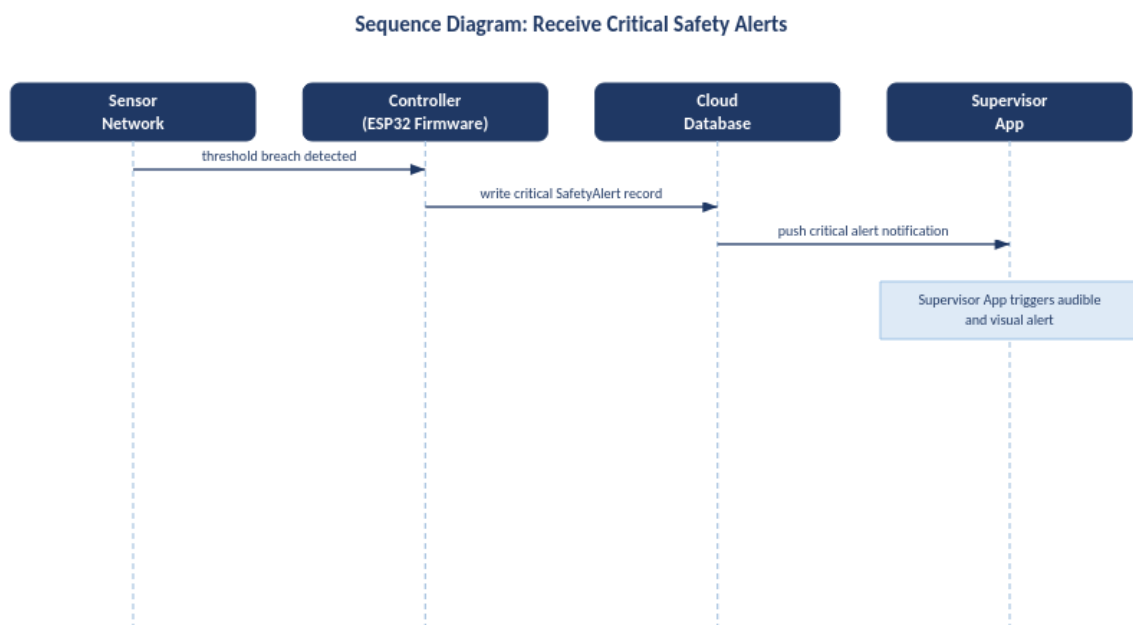


Figure 3.10: Sequence diagram: Receive Critical Safety Alerts

When a sensor threshold breach is detected, the controller writes a critical safety alert which the cloud database pushes to the Supervisor App, triggering an audible and visual alert.

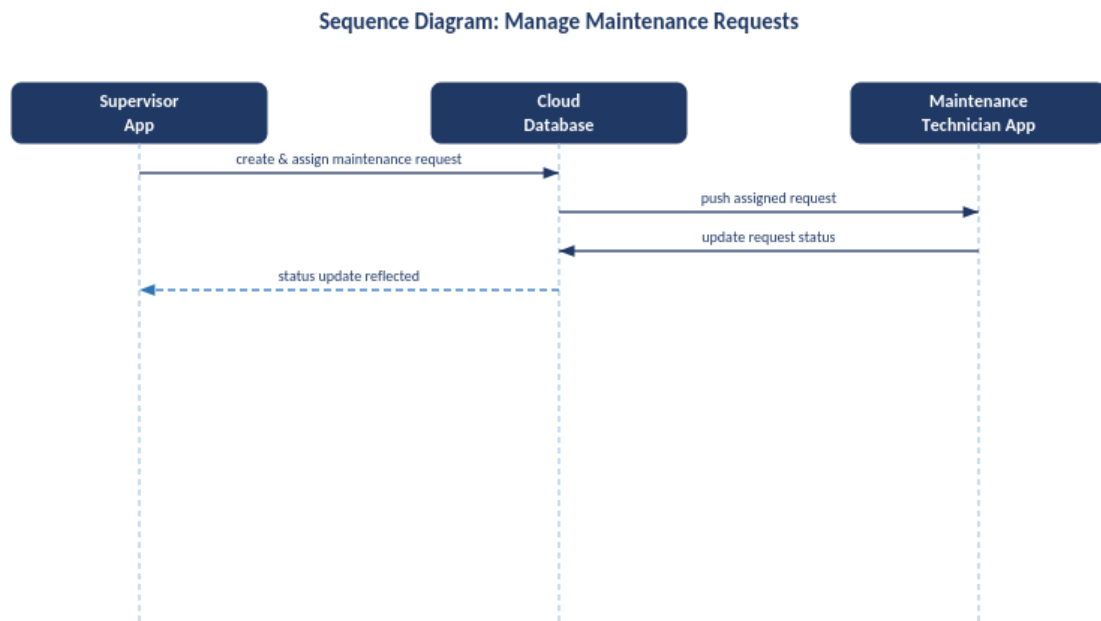


Figure 3.11: Sequence diagram: Manage Maintenance Requests

The Supervisor App creates and assigns a maintenance request; the assigned technician receives it and updates its status, which is reflected back to the Supervisor App.

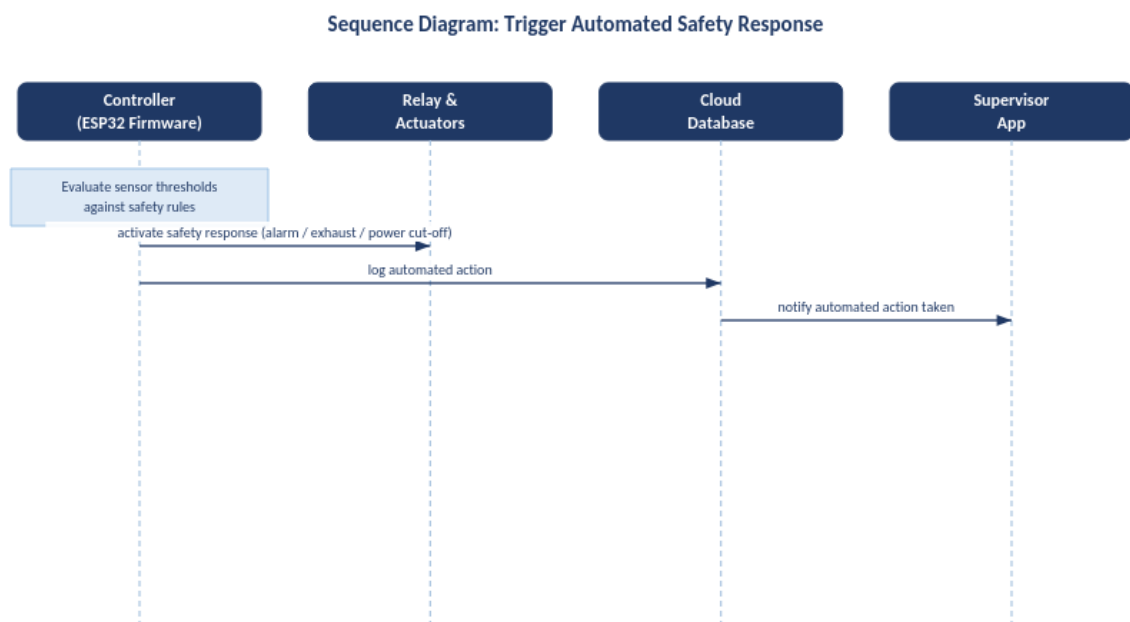


Figure 3.12: Sequence diagram: Trigger Automated Safety Response

The controller evaluates sensor thresholds on-device and, where safety rules are breached, activates the relay-controlled actuators directly while logging the automated action to the cloud database.

### 3.2.5 Context Diagram of Proposed System

Places the system at the centre of all data flow: sensor and QR inputs flow in, while relay-controlled actuators, the worker app, and the supervisor/admin app receive outputs, with a cloud database syncing all activity in real time.

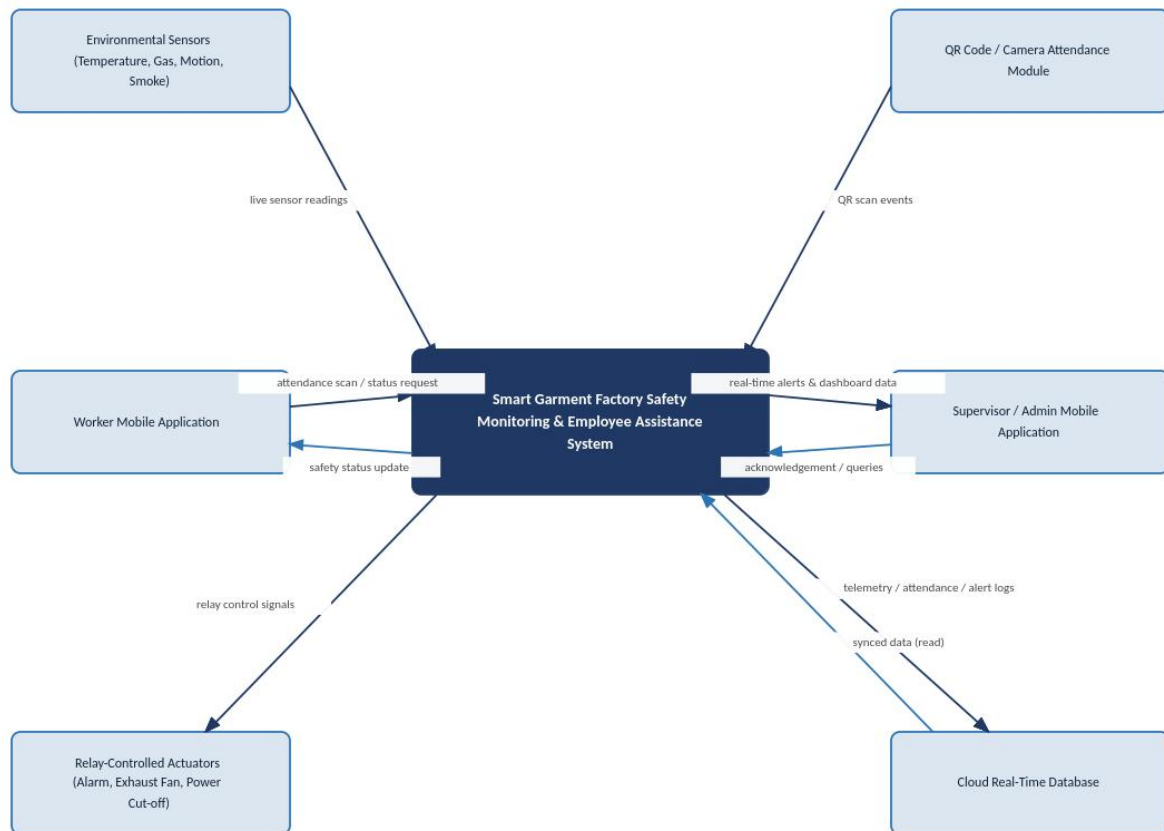


Figure 3.13: System context diagram

### 3.3 ER Diagram of Proposed System

The entity-relationship diagram below shows how attendance, fault, sensor, alert, maintenance and emergency-report data are structured and related within the underlying database, including the blockchain ledger used to keep attendance records tamper-evident.

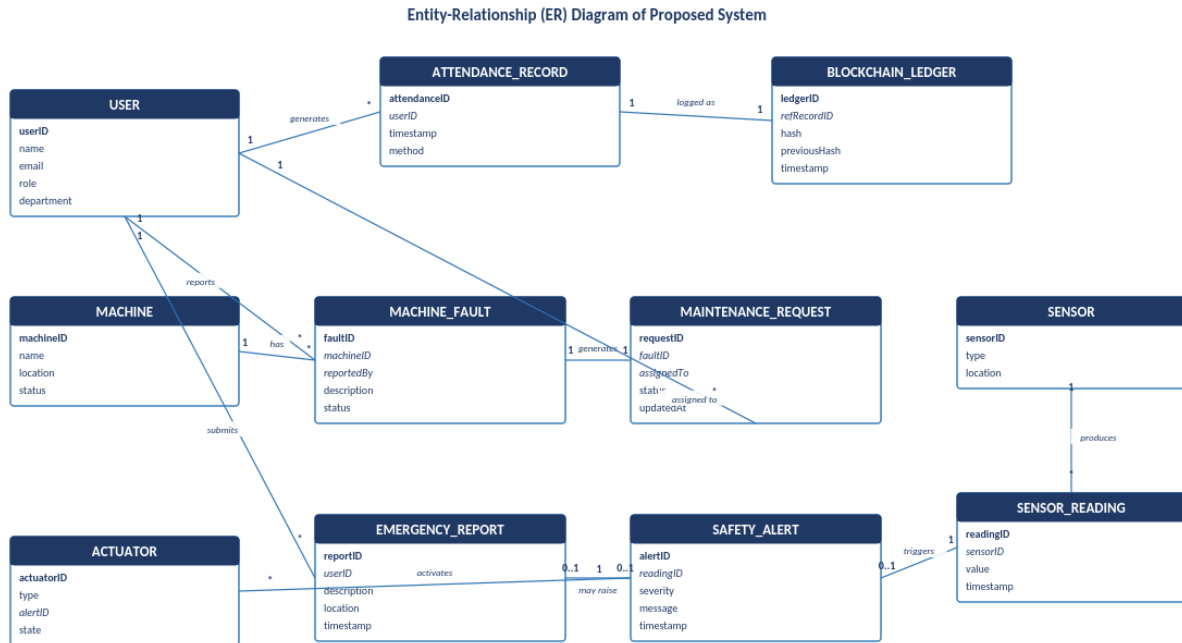


Figure 3.14: ER diagram of proposed system

### **3.4 Chapter Summary**

Through UML modelling grounded in Marsylka Manufacturing Lanka's actual reporting structure and safety technology, this chapter mapped the data flow between physical hazard triggers, automated hardware response, and software-based supervisor notification. The use case diagram clarified how each factory role interacts with the system, the context diagram showed how the system unifies previously disconnected safety devices, and the sequence diagram confirmed that life-safety actions are never gated on network or database availability, ensuring the fire, gas, and machine-fault protocols integrate reliably with the attendance and dashboard features described in later chapters.

## Chapter 4: Solution Design

### 4.1 Introduction

This chapter presents the design of the proposed solution, the **TexyShield** mobile application (branded under the Smart Garment Factory Safety Monitoring & Employee Assistance System), covering its interface design, underlying database structure, and report layouts. Each interface below reflects an implemented screen of the application, described with its purpose and represented through pseudo code that models its core logic. The database design translates the data these interfaces capture and display into a structured schema, and the report layout section documents the printable/exportable outputs the system generates.

## 4.2 Interface Design

### 4.2.1 Splash Screen 1 [App Launch]

Initial launch screen featuring a clean white background with a neon green factory logo ("SMART GARMENT FACTORY APP"). It serves as the primary visual introduction to the application.

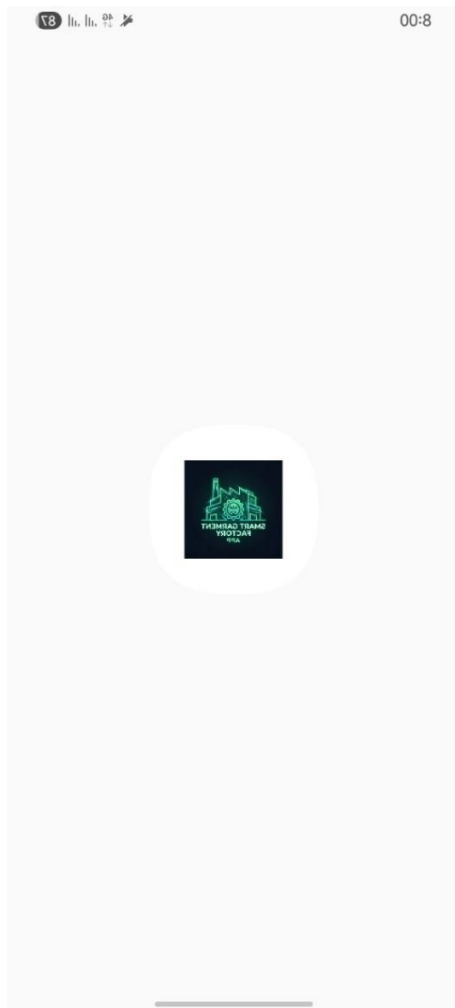


Figure 4.1: Splash Screen 1

BEGIN

Display White Background

Render "Smart Garment Factory" Neon Logo  
at Center

Start Initialization

Wait for minimum duration

Transition to SplashScreen 2

END



#### 4.2.2 Splash Screen 2 [Branding / Auth Check]

Secondary branding splash screen showcasing the "TaxyShield" identity with a dark mode aesthetic. Features a red and white shield emblem, subtitle "Industry 4.0 Safety & Monitoring", and tagline "POWERED BY AI & BLOCKCHAIN".

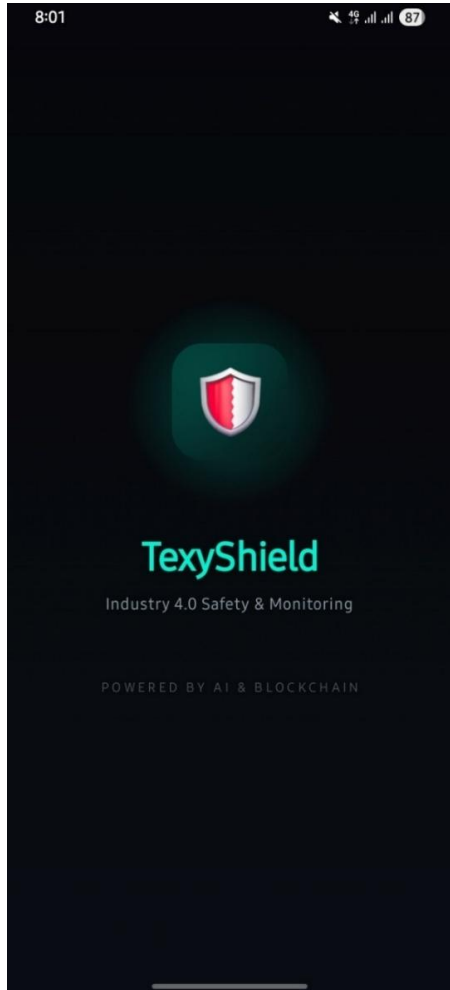


Figure 4.2: Splash Screen 2

BEGIN

Display Dark Background

Render Shield Logo at Center

Render "TaxyShield" Text

Initialize AI & Blockchain Models in  
Background

Wait for minimum duration

IF User is Logged In:

Fetch User Role

IF Role == "Admin":

Navigate to Admin Dashboard

ELSE:

Navigate to Worker Dashboard

ELSE:

Navigate to LandingPage

END

### 4.2.3 Landing Page

Gateway for unauthenticated users, providing navigation paths to log in or register. Clean, welcoming interface with primary call-to-action buttons.

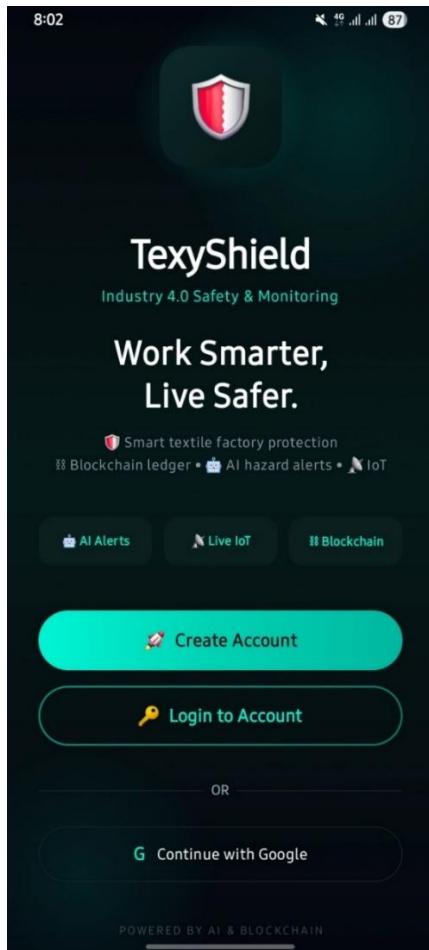


Figure 4.3: Landing Page

BEGIN

Render App Logo

Render "Login" Button

Render "Register" Button

ON "Login" Button Click:

Navigate to LoginScreen

ON "Register" Button Click:

Navigate to RegisterPage

END

#### 4.2.4 Login Screen

Secure authentication interface requiring the user's registered email and password. Includes form validation and visual feedback.

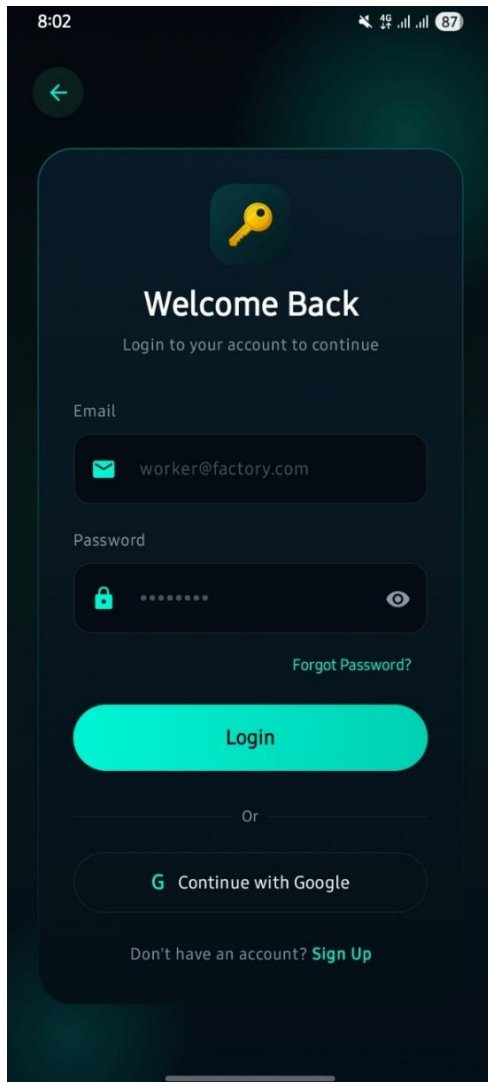


Figure 4.4: Login Screen

BEGIN

Render Email & Password Inputs

Render Login Button

ON Login Button Click:

Display Loading Spinner

Call Authentication API

Hide Loading Spinner

IF Auth Success:

Navigate to Dashboard based on Role

ELSE:

Display Error Alert ("Invalid Credentials")

END

#### 4.2.5 Register Screen [Sign Up]

Registration form designed to capture necessary user details to provision a new account, including role selection.

The image shows a mobile application interface for a registration screen. The screen has a dark background with a light blue header bar. At the top, there is a back arrow icon. Below the header, there is a logo of a building with a red roof. The title 'Sign Up' is displayed in white, followed by the subtitle 'Create an account to continue'. The form consists of three input fields: 'Full Name' with the placeholder 'Alex Lee', 'Email' with the placeholder 'example@email.com', and 'Set Password' with the placeholder 'Alex Lee'. Below the 'Set Password' field, there is a 'Forgot Password?' link. A large blue 'Sign Up' button is positioned below the form. Below the button, there is a horizontal line with the word 'Or' in the center. Below the line, there are two buttons: 'Continue with Google' and 'Continue with Apple'. At the bottom, there is a link 'Already have an account? Log In'.

Figure 4.5: Register Screen

BEGIN

Render Name, Email, Password Inputs

Render Role Selection Dropdown  
(Admin/Worker)

Render Register Button

ON Register Button Click:

Validate Form Inputs

Display Loading Spinner

Call Registration API

Hide Loading Spinner

IF Success:

Display Success Alert

Navigate to LoginScreen

ELSE:

Display Error Alert ("Registration Failed")

END

#### 4.2.6 Admin Dashboard

Command centre tailored for administrators. Features metric cards for Live IoT Sensors (Temperature, Fabric Dust Gas, Current, Motion, Light) and AI Predictions (Fabric Storage Fire Risk, Sewing Machine Overcurrent), plus quick links like Safety Ledger, Attendance, and PPE Detection.

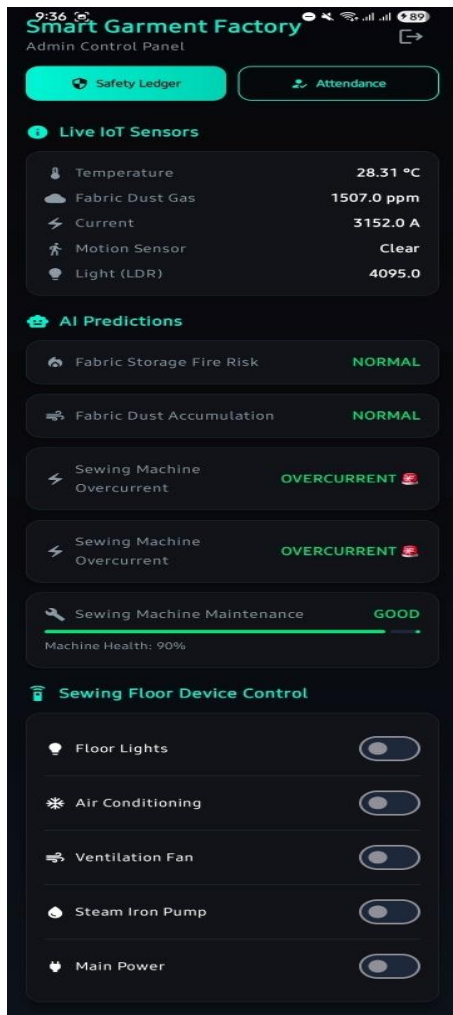


Figure 4.6: Admin Dashboard

BEGIN

Fetch Live IoT Sensors Data

Fetch AI Predictions Status

Render Navigation Buttons (Safety Ledger,  
Attendance, PPE Check)

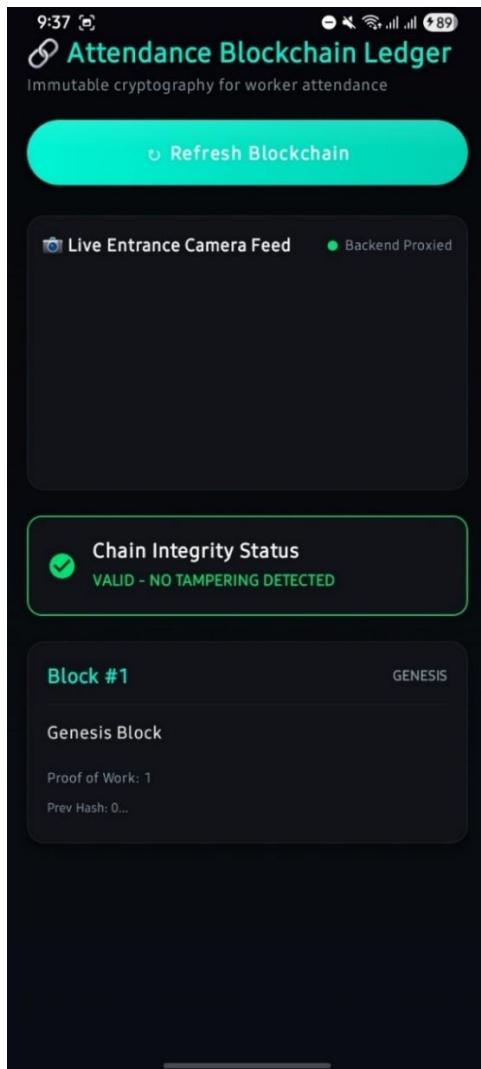
Render Live Data Cards

Render Device Control Toggles

END

#### 4.2.6.1 Attendance Blockchain Ledger Screen

Administrative view displaying a blockchain-backed ledger for worker attendance. Features a Live Entrance Camera Feed, Chain Integrity Status (VALID - NO TAMPERING DETECTED), and chronological blocks (Genesis Block).



BEGIN

Render "Refresh Blockchain" Button

Render "Live Entrance Camera Feed"

Render "Chain Integrity Status"

Display Loading Spinner

Call Blockchain API to fetch attendance blocks

Hide Loading Spinner

Render ListView of Blocks

ON Refresh Click:

Reload Blocks & Verify Integrity

END

Figure 4.7: Attendance Blockchain Ledger

#### 4.2.6.2 Blockchain Safety Ledger Screen

High-security auditing interface accessed via the "Safety Ledger". Presents immutable event logs anchored to a blockchain for critical safety reports.

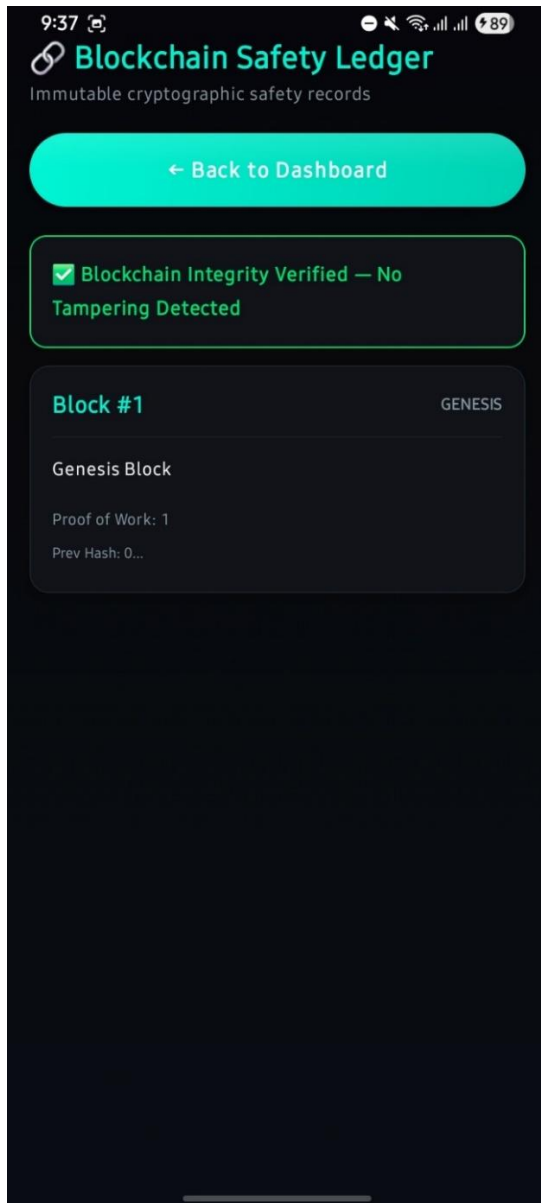


Figure 4.8: Blockchain Safety Ledger

BEGIN

Render Screen Title ("Safety Ledger")

Display Loading Spinner

Call Blockchain API to fetch safety blocks

Hide Loading Spinner

Render ListView of Safety Event Blocks

END

#### 4.2.6.3 Emergency Report Alert in Admin Dashboard

Admin interface for viewing and managing emergency reports submitted by workers. Displays incident details, severity, and location.

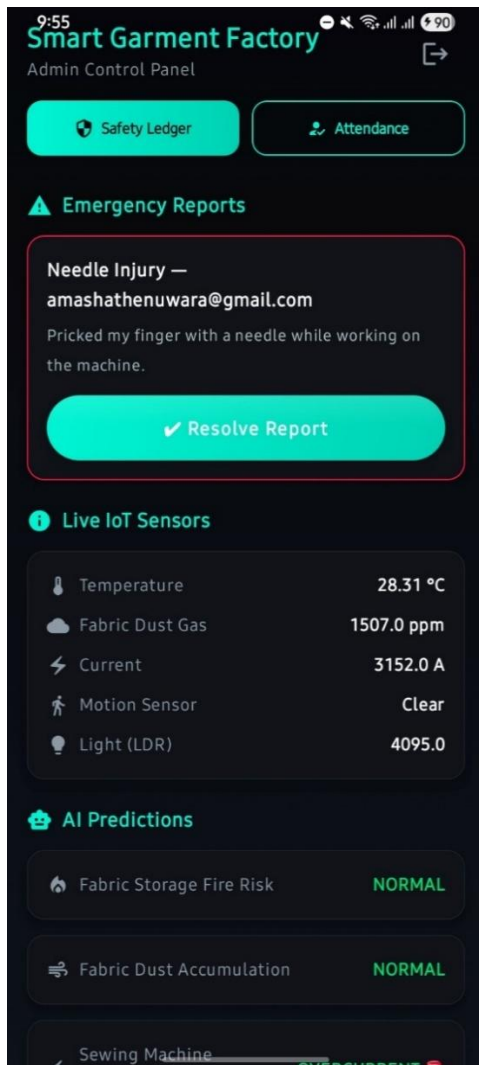


Figure 4.9: Emergency Report Alert (Admin view)

BEGIN

Fetch Active Emergency Reports

Render List of Reports with Severity Tags

ON Report Click:

Open Details and Action Modal

ON Modal Action "Resolve" Click:

Update Report Status to "Resolved"

Close Modal

Refresh Reports List

END



#### 4.2.7 Worker Dashboard

Streamlined interface for factory workers prioritising actionable information such as daily tasks, real-time personal health metrics, machinery scanning, and emergency reporting.

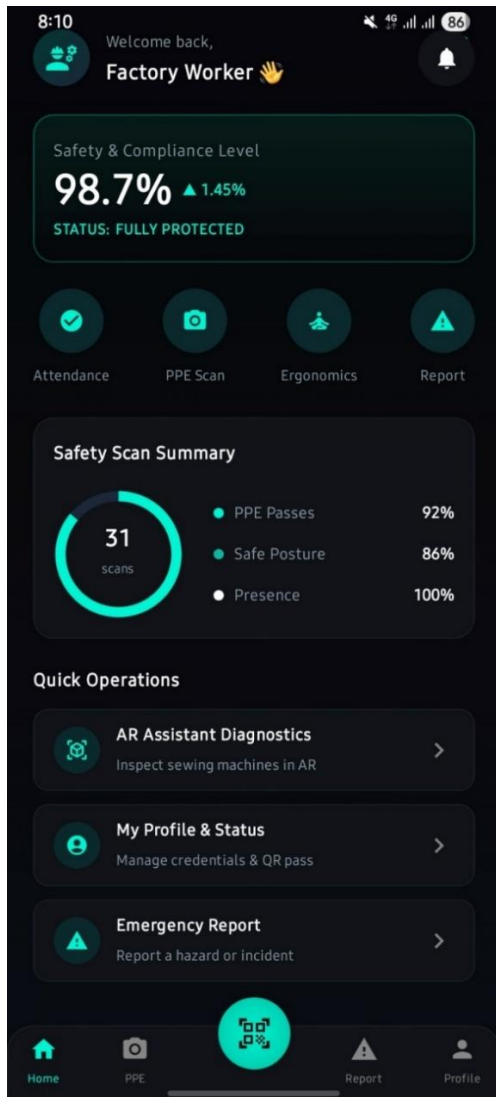


Figure 4.10: Worker Dashboard

BEGIN

Fetch Worker Tasks

Render Health Widget Summary

Render "Scan Machinery (AR)" Button

Render Floating Action Button (FAB) for  
Emergency (SOS)

END

#### 4.2.7.1 Profile Screen

User management interface viewing account details, settings, and sign-out options.

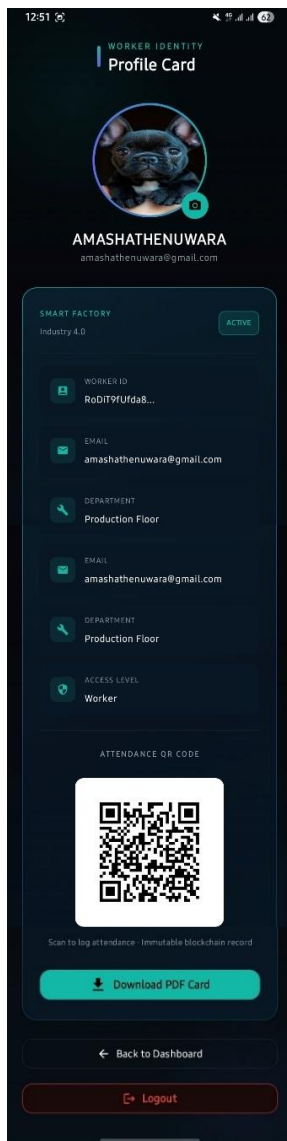


Figure 4.11: Profile Screen

BEGIN

Fetch User Profile Data

Render Profile Details (Name, Email, Role)

Render "Logout" Button

ON Logout Click:

Clear Local Auth Tokens

Navigate to LandingPage

END

#### 4.2.7.2 AR Machine Details Screen

Augmented reality module overlaying digital instructions, maintenance markers, and safety warnings directly onto physical factory machinery.



Figure 4.12: AR Machine Details screen

BEGIN

Request Camera Permissions

Start Camera Preview

Run Machine Recognition Model

IF Machine Recognized:

Fetch Machine Maintenance Data

Render AR Overlays at Anchor Points

ELSE:

Display "Searching for Machinery..." Overlay

END

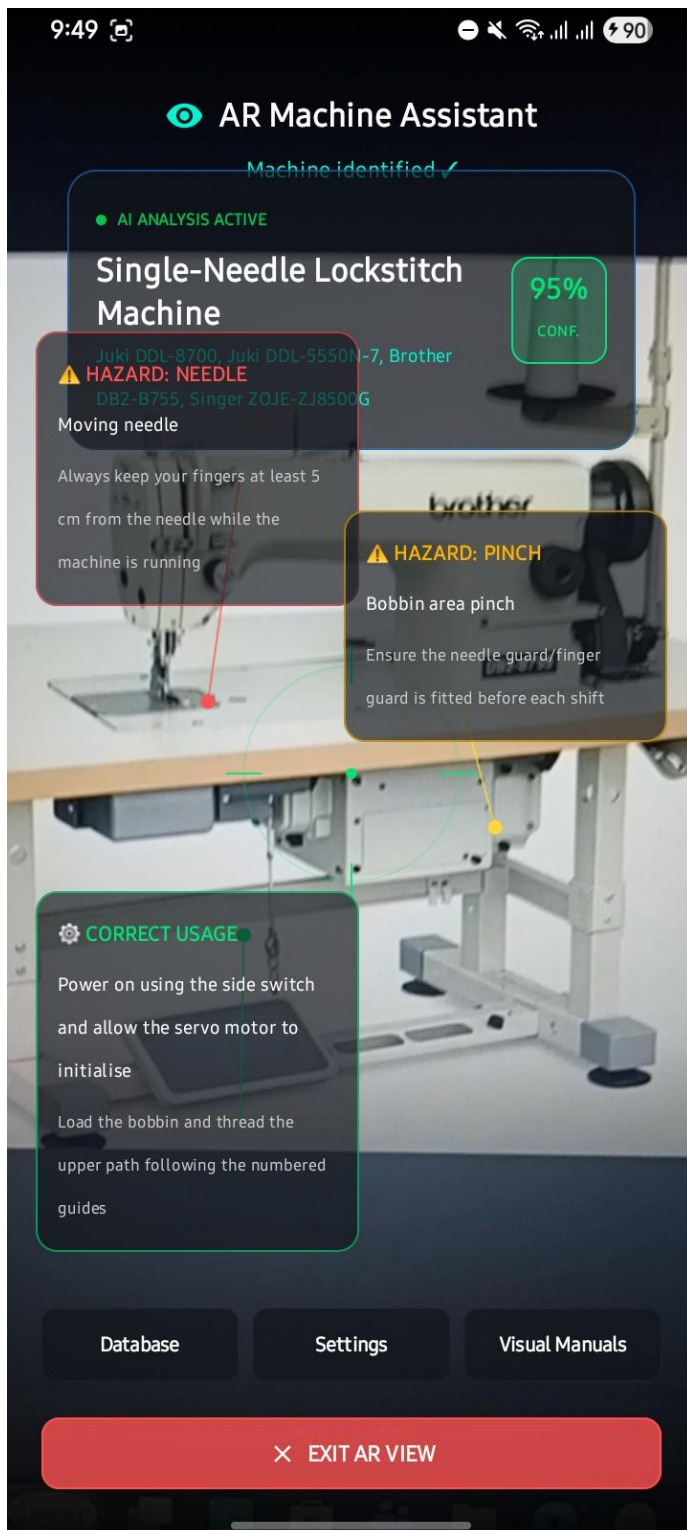


Figure 4.13: AR Overlay in use

#### 4.2.7.3 Emergency Report

Highly accessible for rapid incident reporting by workers. Includes quick-select categories, description, and submit button.

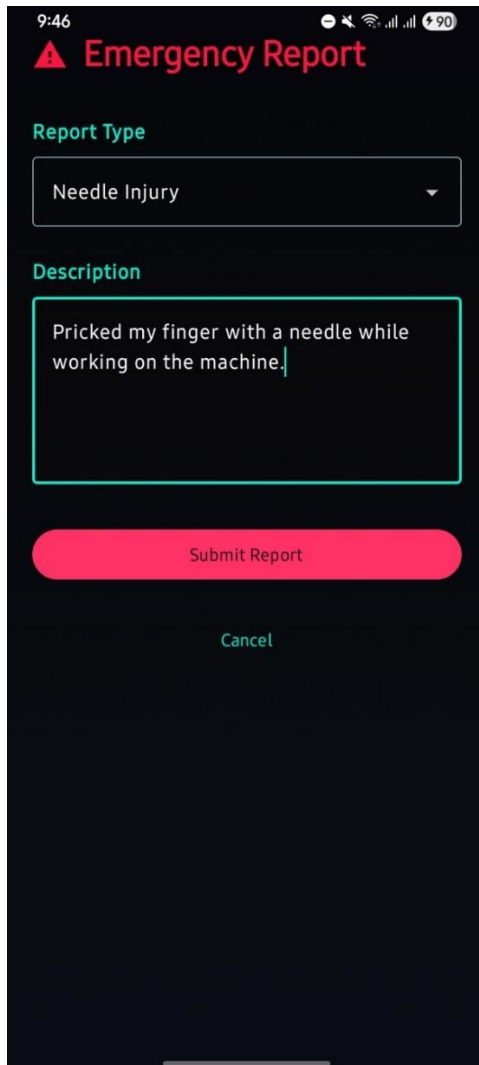


Figure 4.14: Emergency Report screen

BEGIN

Render Emergency Category Grid (Fire, Medical, Machine Failure, etc.)

Render Text Area for Optional Details

Render "SUBMIT ALERT" Button

ON Submit:

Display Loading Spinner

Call Emergency Report API (Include Worker ID & Location)

Hide Loading Spinner

IF Success:

Display Success Alert ("Alert Broadcasted")

Navigate to Worker Dashboard

ELSE:

Display Error Alert

END

#### 4.2.7.4 PPE Detection [Entrance]

AI-powered computer vision screen analysing camera feed to detect mandatory PPE (face mask, gloves). Flashes green if detected, highlights in red if missing.

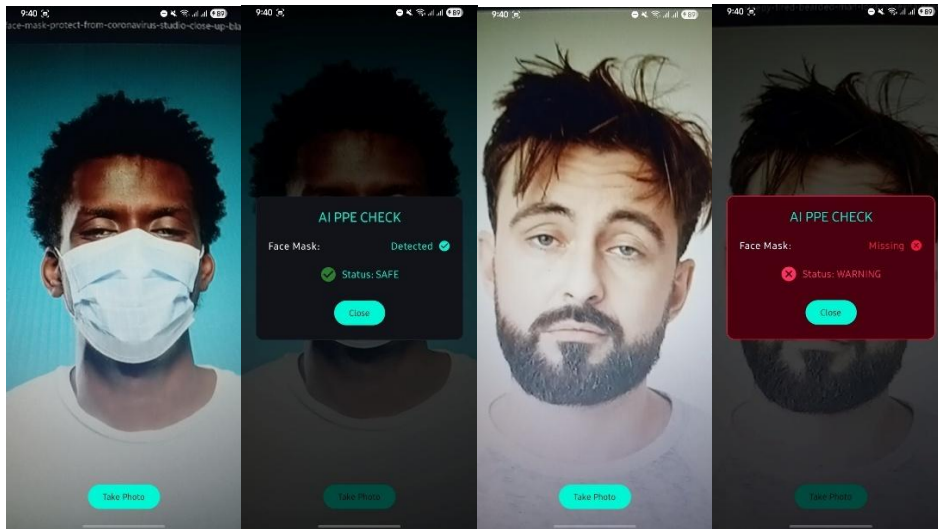


Figure 4.15: PPE Detection screens

BEGIN

Request Camera Permissions

Start Camera Preview

Run PPE Object Detection Model on Frame Stream

IF PPE Detected (Confidence > Threshold):

Display "Compliance Confirmed" Overlay (Green)

Log Event to Database

ELSE:

Display "Missing Equipment: [Item Name]" Overlay (Red)

Log Warning Event

END

#### 4.2.7.5 WorkerHealthDetection Screen

Health monitoring dashboard visualising telemetry data from wearable devices (Heart Rate, Body Temperature) to monitor worker fatigue.

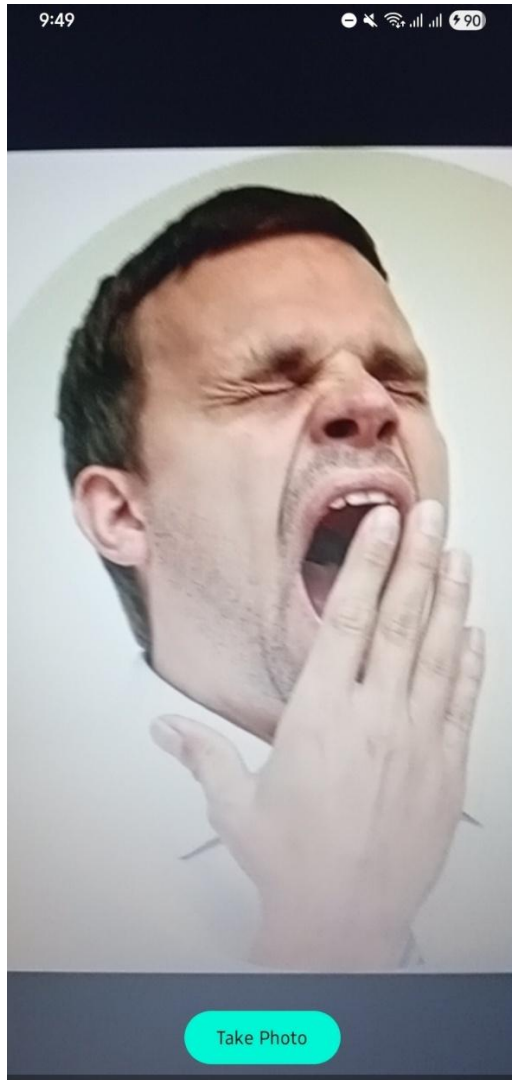


Figure 4.16: Worker Health Detection screen

BEGIN

Subscribe to Health Data WebSocket/Stream

Render Real-time Line Charts (HR & Temp)

ON Data Update:

Update Charts

IF Data > Safety Threshold:

Render Warning Overlay ("High Heart Rate Detected. Please take a break.")

Trigger Local Notification

END

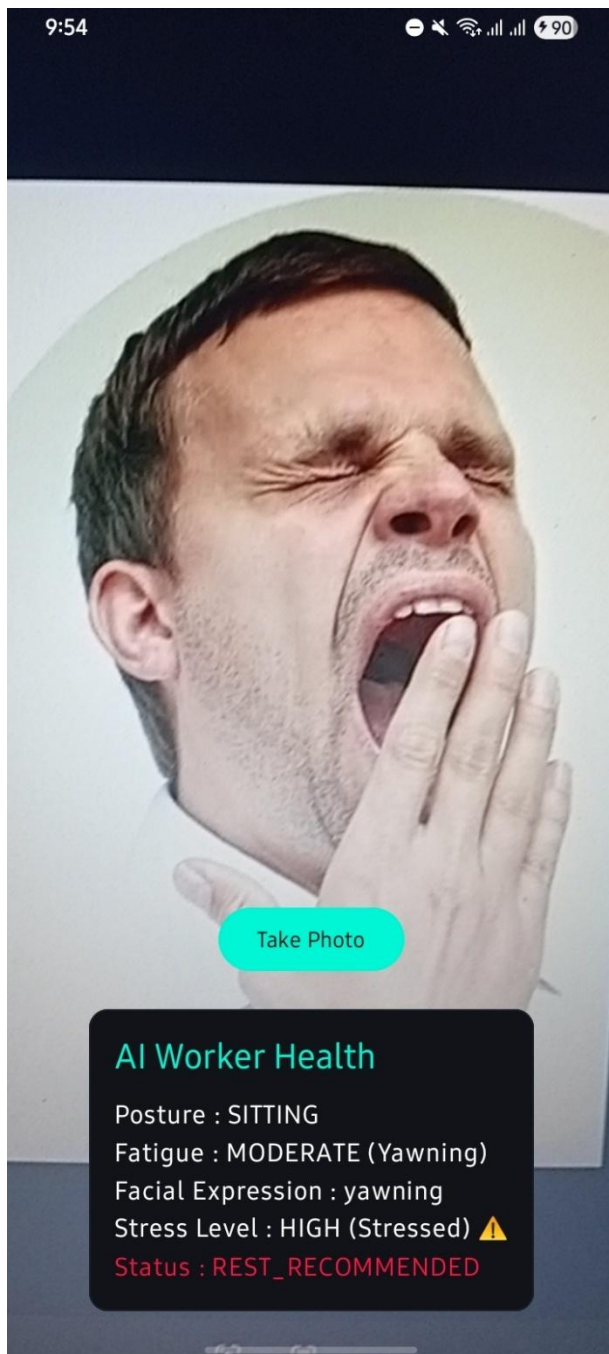


Figure 4.17: Health trend view



## 4.3 Database Schema & Architecture details

### 4.3.1 Overview

The TaxyShield Smart Garment Factory system uses Firebase Realtime Database (RTDB) as its primary data store. Unlike a relational database, which organises records into fixed rows, columns and tables, or Firestore, which organises data into discrete collections and documents, the Realtime Database stores all application data as a single, large JSON tree. Every entity in the system attendance records, user profiles, sensor readings, emergency reports, health metrics, and PPE compliance logs exists as a nested node within this one tree, and the entire structure is synchronised across connected clients in real time.

This architecture was selected because the core requirement of the system is low-latency, bidirectional data streaming between factory-floor devices (IoT sensors, wearables, entrance cameras) and the mobile/web dashboards used by workers and administrators, rather than complex relational querying.

### 4.3.2 Justification for Realtime Database

**Live synchronisation:** Every client subscribed to a node receives updates within milliseconds of a change, which is essential for live machine telemetry, health monitoring and emergency alerts.

**Offline-first behaviour:** The Firebase SDK maintains a local, in-memory/disk cache on the device. If the factory floor loses connectivity, the app continues to read and write locally (for example, queuing an Attendance check-in or an Emergency Report) and automatically synchronises those queued writes with the cloud JSON tree once the connection is restored.

**Simple, predictable structure:** A single JSON tree keeps the data model easy to reason about for a system with a moderate number of well-defined entities, and avoids the overhead of managing multiple collections and composite indexes.

**Low operational overhead:** RTDB requires minimal server-side configuration, which suited the project timeline and team size.

### 4.3.3 Complete JSON Tree Structure

Below is the full hierarchical structure of the database. Each top-level key represents a root node; nested keys prefixed with "\$" (such as \$userId, \$machineId, or \$cameraId) represent dynamic identifiers rather than literal keys, and \$pushId denotes a Firebase-generated unique record identifier.

```
{
  "Attendance": {
    "$userId": {
      "$pushId": {
        "date": "2026-07-14",
        "status": "Present",
        "time": "1784380977737",
        "userId": "RoDiT9fUfda85jR6riPXaGS11M23"
      }
    }
  },

  "Users": {
    "$userId": {
      "fullName": "John Doe",
      "email": "worker@factory.com",
      "role": "Worker",
      "isActive": true,
      "createdAt": "1784380900000"
    }
  },

  "SensorTelemetry": {
    "$machineId": {
      "$pushId": {
        "sensorType": "Temperature",
        "value": 45.5,
        "aiRiskFlag": false,
        "time": "1784380977737"
      }
    }
  }
}
```

},

"EmergencyReports": {

"\$pushId": {

"workerId": "RoDiT9fUfda85jR6riPXaGS11M23",

"category": "Machine Failure",

"description": "Sewing machine motor overheating",

"severity": "High",

"status": "Pending",

"time": "1784380977737"

}

},

"WorkerHealth": {

"\$userId": {

"\$pushId": {

"heartRate": 85,

"bodyTemp": 36.6,

"fatigueWarning": false,

"time": "1784380977737"

}

}

},

"PPELogs": {

"\$cameraId": {

"\$pushId": {

"workerId": "RoDiT9fUfda85jR6riPXaGS11M23",

"status": "Missing PPE",

"missingItems": "Hard Hat",

"time": "1784380977737"

}

}

}

}

#### 4.3.4 Node Descriptions

The table below summarises the purpose and nesting strategy of each root node in the database. Nodes that are frequently queried per-entity (Attendance, SensorTelemetry, WorkerHealth, PPELogs) are grouped under the relevant identifier so the client can attach a listener to a single branch rather than the whole tree, which reduces bandwidth and improves read performance. EmergencyReports is deliberately kept flat so administrators can retrieve a single, factory-wide list of active cases.

*Table 4.1: Firebase Realtime Database root node descriptions*

| Root Node        | Nesting Structure                     | Description                                                                                                                                                                                                                                                                                   |
|------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attendance       | /Attendance/\$userId/\$pushId         | Records daily worker attendance. Entries are grouped under each worker's unique <code>userId</code> , allowing the app to efficiently query a single employee's attendance history without scanning the entire node. A Firebase push ID is generated for every new check-in/check-out record. |
| Users            | /Users/\$userId                       | Stores employee profile information and links each Firebase Authentication UID ( <code>userId</code> ) to the corresponding factory role (Admin or Worker), full name, email, and account status.                                                                                             |
| SensorTelemetry  | /SensorTelemetry/\$machineId/\$pushId | Captures real-time IoT sensor readings (e.g. temperature) from factory machinery. Data is grouped by <code>machineId</code> so dashboards can subscribe to live updates for a specific machine, and includes an AI-generated risk flag for anomaly detection.                                 |
| EmergencyReports | /EmergencyReports/\$pushId            | Maintains a flat, ungrouped list of all emergency reports raised by workers. Because it is not nested by user, administrators can retrieve a single                                                                                                                                           |

|              |                                 |                                                                                                                                                                                                     |
|--------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                                 | consolidated list of pending or active emergencies across the entire factory floor.                                                                                                                 |
| WorkerHealth | /WorkerHealth/\$userId/\$pushId | Logs live biometric data streamed from wearable devices, including heart rate, body temperature, and a fatigue-warning flag. Grouping by userId enables continuous, per-worker health monitoring.   |
| PPELogs      | /PPELogs/\$cameraId/\$pushId    | Stores Personal Protective Equipment (PPE) compliance results generated by AI-powered entrance cameras. Records are grouped by cameraId to trace detections back to a specific monitoring location. |

#### 4.3.5 Key Design Considerations

##### 4.3.5.1 Data Grouping and Query Efficiency

Nodes that require per-entity history (Attendance, SensorTelemetry, WorkerHealth, PPELogs) are indexed first by the relevant identifier (userId, machineId, or cameraId) and then by a push ID for each event. This two-level nesting allows the application to attach a Firebase listener directly to a specific worker's or machine's branch, avoiding the cost of filtering the entire dataset on the client.

##### 4.3.5.2 Offline Synchronisation

Firebase Realtime Database's built-in disk persistence allows the mobile application to remain functional without network access. Writes made offline, such as an attendance check-in or a newly filed emergency report, are held in a local write-ahead queue and automatically replayed against the cloud database as soon as connectivity resumes, ensuring no data loss on the factory floor.

##### 4.3.5.3 Timestamps

All time-based fields (time, createdAt) are stored as Unix epoch millisecond strings rather than formatted date strings. This keeps values sortable and timezone-independent, and allows the client to format them for display according to the user's locale.

##### 4.3.5.4 Security Rules

Access to each root node is restricted through Firebase Security Rules based on the authenticated user's UID and role (Admin or Worker), so that a worker can only read/write their own Attendance, WorkerHealth and EmergencyReports entries, while Admin accounts are granted read access across all nodes for monitoring and reporting.

## 4.4 Artificial Intelligence Model Training & Methodology

The TaxyShield system utilises a dedicated Python FastAPI backend (ai-service/) to process sensor data and run advanced machine learning models. The system relies on two primary classes of machine learning: Deep Learning (Computer Vision) and Ensemble Learning (Sensor Telemetry).

### 4.4.1 Frameworks and Training Process

The computer vision models were trained using the Ultralytics YOLO framework, specifically leveraging YOLOv8 and YOLO11 architectures for object detection and image classification. Custom datasets were curated and annotated, including images of workers with and without masks, various sewing machines, and faces exhibiting signs of fatigue. The models were trained using PyTorch on GPUs. The resulting PyTorch weight files (.pt) are dynamically loaded by the FastAPI server to run real-time inferences on incoming video streams from the factory floor.

### 4.4.2 Safety & PPE Detection Models

Purpose: To detect if workers are wearing required face masks or gloves.

Methodology: The system uses YOLOv8, a state-of-the-art Convolutional Neural Network (CNN). Unlike older two-stage models, YOLO passes the entire image through a single neural network, predicting bounding boxes and class probabilities simultaneously. This single-pass architecture allows it to run at extremely high frame rates, which is critical for processing live CCTV streams where instantaneous alerts are required to prevent accidents.

### 4.4.3 Worker Health & Ergonomics Models

Purpose: To analyse facial expressions for fatigue and stress, and to detect if a worker has collapsed (fall detection).

Methodology: This module utilises YOLO Image Classification and YOLO-Pose (Pose Estimation). A specialised YOLO-Classification head was trained on facial micro-expressions to categorise the worker's state (e.g., Alert, Fatigued, Stressed). For fall detection, YOLO-Pose extracts human skeletal keypoints (joints, spine) and tracks their coordinates in real-time. Sudden, drastic changes in the vertical alignment of the spine trigger an immediate fall alert. This

approach provides sub-millisecond inference times, allowing the system to track multiple workers on the floor simultaneously.

#### 4.4.4 Machine Identification Models

**Purpose:** To identify the specific type of sewing machine from a camera feed and load the correct Augmented Reality (AR) overlays for mechanics.

**Methodology:** A YOLOv8 Classification Model was trained using transfer learning on thousands of augmented images of different sewing machines. Those images were obtained from Marsylka Manufacturing Lanka (Pvt) Ltd garment factory[client]. This deep CNN abstracts robust structural features, allowing the AR camera to instantly recognise the machine type regardless of the highly variable lighting and occlusions typical in a factory environment.



## Chapter 5: Conclusion

Built using Marsylka Manufacturing Lanka (Pvt) Ltd, a 750-employee, 14-line ladies' wear manufacturer as the case study, the TaxyShield system replaced the factory's reactive maintenance and fragmented manual safety practices with unified real-time IoT sensing, AI risk prediction, QR attendance, blockchain records, and a mobile app for workers and admins.

All core objectives were met: real-time monitoring, automated hazard response, contactless attendance, centralised visibility, digital fault reporting, and worker wellbeing support.

As an academic project based on a single case study, the AI models need more data to generalise, and the blockchain/AR modules remain in their early stages. Future work includes expanding training data, integrating existing CCTV, and piloting on a few lines before a full rollout.

Overall, the system strengthens rather than replaces Marsylka's existing safety policies, offering a scalable path to safer ladies' wear manufacturing.

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## Appendices

### Approval Letter from Organization

**MARSYLKA®**  
**MARSYLKA MANUFACTURING LANKA (PVT) LTD**

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22<sup>nd</sup> July 2026

To Whom It May Concern,

**Subject: Approval to Conduct HND Final Year Academic Project**

This letter is to confirm that Ms. Amasha Thenuwara, an undergraduate student following the Higher National Diploma (HND) in Software Engineering at the National Institute of Business Management (NIBM), has been granted permission to conduct the academic project titled:

***"Smart AI-Based Garment Factory Safety Monitoring and Employee Assistance System."***

We understand that this project is being carried out as part of the HND final year academic requirements.

Our organization grants permission for the student to observe relevant workplace activities and collect the necessary information required for the successful completion of the project. This may include taking photographs of the factory environment and equipment, with prior approval, solely for academic and research purposes.

The student has agreed that any confidential information obtained during the project will be treated as strictly confidential and will not be disclosed or used for any purpose other than the academic project without our permission.

We wish the student every success in completing this project.

Yours faithfully,

  
**Marsylka Manufacturing Lanka (Pvt) Ltd**  
**Human Resources Manager**  
Human Resources Manager

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Head Office & Factory 1: Block-10, Export Processing Zone, Polgahawela, Sri Lanka, Tel+94 37 494 4371-3 Fax: +94 37 494 4374  
Factory 2: Heraliyawala Industrial Zone, Heraliyawala, Kurumegala Tel+94 37 222 3678  
E-Mail : [info@marsylka.lk](mailto:info@marsylka.lk) Web: [www.marsylka.com](http://www.marsylka.com) REG. NO. P.V.8383

Figure 5.1: Approval Letter from Marsylka Manufacturing Lanka (Pvt) Ltd

## Questioners and Interview Questions

As part of the data-gathering process for the HND Final Year Project titled “Smart AI-Based Garment Factory Safety Monitoring and Employee Assistance System,” a structured questionnaire was distributed to an industry representative to collect first-hand information on factory operations, machine management, workplace safety practices, and the perceived value of an AI-driven monitoring solution. The survey was completed by Mrs R.P.A.P. Jayarathne, Dedicated Quality Assurance Officer at Marsylka Manufacturing Lanka (Pvt) Ltd, a garment manufacturing facility located in the Export Processing Zone, Polgahawela. The responses summarised below provide the real-world context and requirements that informed the design of the system, including its safety monitoring, PPE detection, and emergency reporting features.



*Figure 5.2: Survey QR code*

*Scan to view the original Google Form*

### Survey Response Summary

*Table 5.1: Survey Response Summary*

|                                        |                                        |
|----------------------------------------|----------------------------------------|
| <b>Respondent's Name</b>               | R.P.A.P. Jayarathne                    |
| <b>Position</b>                        | Dedicated Quality Assurance            |
| <b>Factory Name</b>                    | Marsylka Manufacturing Lanka (Pvt) Ltd |
| <b>Factory Location</b>                | Export Processing Zone, Polgahawela    |
| <b>Types of Garments Manufactured</b>  | Ladies Wears                           |
| <b>Approximate Number of Employees</b> | 750                                    |
| <b>Number of Production Lines</b>      | 14                                     |

|                                                               |                                                                                                                                     |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Types of Sewing Machines Used</b>                          | Lockstitch, Overlock, Flatlock, Buttonhole, Button Sewing, Bartack, Zigzag                                                          |
| <b>Approx. Number of Sewing Machines in Operation</b>         | 550                                                                                                                                 |
| <b>Machine Maintenance Frequency</b>                          | Weekly                                                                                                                              |
| <b>Common Sewing Machine Problems</b>                         | Thread Breakage, Needle Breakage, Fabric Jamming, Motor Failure, Machine Overheating, Noise/Vibration, Electrical Problems          |
| <b>How Faults Are Reported</b>                                | Inform Supervisor, Maintenance Team                                                                                                 |
| <b>Do Breakdowns Affect Daily Production?</b>                 | Yes                                                                                                                                 |
| <b>Workplace Safety Policy in Place?</b>                      | Yes                                                                                                                                 |
| <b>Required PPE for Workers</b>                               | Face Mask, Gloves                                                                                                                   |
| <b>Regular Safety Training Conducted?</b>                     | Yes                                                                                                                                 |
| <b>Common Safety Risks</b>                                    | Fire risk, health risk                                                                                                              |
| <b>Emergency Management Approach</b>                          | Following fire procedures and compliance requirements of buyers, maintaining smoke areas, and providing fire training to employees. |
| <b>Digital Safety Monitoring System in Use?</b>               | No                                                                                                                                  |
| <b>Technologies Currently Used</b>                            | CCTV Cameras, Smoke Sensors, Fire Alarm System                                                                                      |
| <b>Challenges in Monitoring Worker Safety</b>                 | Difficult to maintain 100% compliance procedures while achieving maximum production efficiency, which directly affects safety.      |
| <b>AI-Based Safety Monitoring System Beneficial?</b>          | Yes                                                                                                                                 |
| <b>Would Real-Time Safety Alerts Be Useful?</b>               | Yes                                                                                                                                 |
| <b>Would a Mobile App for Supervisors/Managers Be Useful?</b> | Yes                                                                                                                                 |
| <b>Suggestions for Improving Workplace Safety</b>             | Need to improve machine and floor safety, fire and chemical safety, ergonomics, and health.                                         |
| <b>Permission to Use Factory as Case Study?</b>               | Yes                                                                                                                                 |
| <b>Permission to Take Photographs for Academic Use?</b>       | Yes                                                                                                                                 |

## Garment Factory Information Survey

HND Final Year Project

**Project Title:** Smart AI-Based Garment Factory Safety Monitoring and Employee Assistance System

Thank you for taking the time to complete this questionnaire. This survey is conducted as part of my Higher National Diploma (HND) Final Year Project at the National Institute of Business Management (NIBM). Its purpose is to gather information about garment factory operations, workplace safety practices, sewing machine management, and current technologies to support the development of a smart safety monitoring system. Your responses will be used strictly for academic and research purposes, and all information provided will be kept confidential.

### Respondent Information

**Basic details about the person completing this survey.**

**Respondent's Name**

R.P.A.P. Jayarathne

**Position**

Dedicated Quality Assurance

### Factory Information

**General background information about your factory's operations and structure.**

[https://docs.google.com/forms/d/14m8GK4P2Ox7DduM\\_MEt1WGXeCC3PSYd2gV8zzC\\_OQnd8l#responses](https://docs.google.com/forms/d/14m8GK4P2Ox7DduM_MEt1WGXeCC3PSYd2gV8zzC_OQnd8l#responses)

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05/08/2026, 22:26

Garment Factory Information Survey

**What types of sewing machines are used in your factory?**

- ☒ Lockstitch Machine
- ☒ Overlock Machine
- ☒ Flatlock Machine
- ☒ Buttonhole Machine
- ☒ Button Sewing Machine
- ☒ Bartack Machine
- ☒ Zigzag Machine

☐ Other: \_\_\_\_\_

**Approximately how many sewing machines are currently in operation?**

550

**How often are sewing machines serviced or maintained?**

- ☐ Weekly
- ☐ Monthly
- ☐ Every 3 Months
- ☒ Every 6 Months
- ☐ Only When a Fault Occurs
- ☐ Other: \_\_\_\_\_

[https://docs.google.com/forms/d/14m8GK4P2Ox7DduM\\_MEt1WGXeCC3PSYd2gV8zzC\\_OQnd8l#responses](https://docs.google.com/forms/d/14m8GK4P2Ox7DduM_MEt1WGXeCC3PSYd2gV8zzC_OQnd8l#responses)

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**Factory Name**

Marsyika Manufacturing Lanka (pvt) Ltd

**Factory Location**

Export Processing Zone, Polgahawela

**What types of garments does your factory manufacture?**

Ladies Night Wears

**Approximately how many employees work in your factory?**

750

**How many production lines are available in your factory?**

14

### Machine Information

**Details about the machinery and equipment used in your factory's production process, including type, condition, and maintenance practices.**

[https://docs.google.com/forms/d/14m8GK4P2Ox7DduM\\_MEt1WGXeCC3PSYd2gV8zzC\\_OQnd8l#responses](https://docs.google.com/forms/d/14m8GK4P2Ox7DduM_MEt1WGXeCC3PSYd2gV8zzC_OQnd8l#responses)

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Garment Factory Information Survey

**What are the most common problems experienced with sewing machines?**

- ☒ Thread Breakage
- ☒ Needle Breakage
- ☐ Fabric Jamming
- ☐ Motor Failure
- ☐ Machine Overheating
- ☐ Noise/Vibration
- ☐ Electrical Problems
- ☐ Other: \_\_\_\_\_

**How are sewing machine faults reported?**

- ☒ Inform Supervisor
- ☐ Maintenance Team
- ☐ Manual Log Book
- ☐ Digital System
- ☐ Other: \_\_\_\_\_

**Do machine breakdowns affect daily production?**

- ☒ Yes
- ☐ No
- ☐ Sometimes

[https://docs.google.com/forms/d/14m8GK4P2Ox7DduM\\_MEt1WGXeCC3PSYd2gV8zzC\\_OQnd8l#responses](https://docs.google.com/forms/d/14m8GK4P2Ox7DduM_MEt1WGXeCC3PSYd2gV8zzC_OQnd8l#responses)

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Garment Factory Information Survey

## Worker Safety

Questions about current safety policies, equipment, and risk management practices.

Does your factory have a workplace safety policy?

- ☒ Yes
- ☐ No

What Personal Protective Equipment (PPE) are workers required to wear?

- ☒ Face Mask
- ☒ Hair Net
- ☒ Gloves
- ☒ Safety Shoes
- ☐ Safety Vest
- ☐ Ear Protection
- ☐ Other: \_\_\_\_\_

Are regular safety training sessions conducted?

- ☒ Yes
- ☐ No

[https://docs.google.com/forms/d/14m8GK4P2Ox7DducM\\_MEE1WGXeGC3PSYd2gVazzc\\_OQiedt#responses](https://docs.google.com/forms/d/14m8GK4P2Ox7DducM_MEE1WGXeGC3PSYd2gVazzc_OQiedt#responses)

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Garment Factory Information Survey

What challenges do you face in monitoring worker safety?

It is difficult to handle 100% compliance procedures while achieving maximum efficiency in production and it will affect directly to the safety.

## Project Feedback

Your opinions on the proposed AI-based safety monitoring system, to help guide the project's design.

Do you think an AI-based safety monitoring system would be beneficial to your factory?

- ☒ Yes
- ☐ No
- ☐ Not Sure

Which features would you like to see in such a system?

Would real-time safety alerts be useful?

- ☒ Yes
- ☐ No

[https://docs.google.com/forms/d/14m8GK4P2Ox7DducM\\_MEE1WGXeGC3PSYd2gVazzc\\_OQiedt#responses](https://docs.google.com/forms/d/14m8GK4P2Ox7DducM_MEE1WGXeGC3PSYd2gVazzc_OQiedt#responses)

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Garment Factory Information Survey

What are the most common safety risks in your factory?

Fire risk, health risk

How are emergencies such as fire, smoke, or gas leaks managed?

Following fire procedures and policies according to compliance requirements of buyers, maintaining smoke areas, giving fire trainings to employees.

## Technology

Details about existing monitoring systems and technology used in your factory.

Does your factory currently use a digital safety monitoring system?

- ☐ Yes
- ☒ No

Which technologies are currently used in your factory?

- ☒ CCTV Cameras
- ☒ Smoke Sensors
- ☒ Temperature Sensors
- ☐ Gas Sensors
- ☒ Fire Alarm System
- ☐ Motion Sensors
- ☐ Other: \_\_\_\_\_

[https://docs.google.com/forms/d/14m8GK4P2Ox7DducM\\_MEE1WGXeGC3PSYd2gVazzc\\_OQiedt#responses](https://docs.google.com/forms/d/14m8GK4P2Ox7DducM_MEE1WGXeGC3PSYd2gVazzc_OQiedt#responses)

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Garment Factory Information Survey

Would a mobile application for supervisors and managers be useful?

- ☒ Yes
- ☐ No

Do you have any suggestions for improving workplace safety?

Need to improve Machine and floor safety, fire and chemical safety, ergonomics and health.

## Permission

Your consent is required to use this factory's information as a case study in this academic project. All data will remain confidential and used solely for academic purposes.

Do you grant permission to use your factory as the case study for this HND Final Year Project?

- ☒ Yes
- ☐ No

Do you permit photographs of the factory environment and equipment to be taken for academic purposes only?

- ☒ Yes
- ☐ No

[https://docs.google.com/forms/d/14m8GK4P2Ox7DducM\\_MEE1WGXeGC3PSYd2gVazzc\\_OQiedt#responses](https://docs.google.com/forms/d/14m8GK4P2Ox7DducM_MEE1WGXeGC3PSYd2gVazzc_OQiedt#responses)

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Garment Factory Information Survey

Additional Comments

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Thank you for your valuable time and support.

This content is neither created nor endorsed by Google.

Google Forms

[https://docs.google.com/forms/d/14n8GK4PZ0c7D8vcM\\_ME1WGXeOC3P5Y62g7kuzC\\_OQ/edit#responses](https://docs.google.com/forms/d/14n8GK4PZ0c7D8vcM_ME1WGXeOC3P5Y62g7kuzC_OQ/edit#responses)

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Figure 5.3: Survey response screenshots



## **Git Hub Link**

[https://github.com/AmashaThenuwara/Industry\\_4.0\\_SmartFactorySafety-MonitoringSystem.git](https://github.com/AmashaThenuwara/Industry_4.0_SmartFactorySafety-MonitoringSystem.git)



# **TexyShield**

Industry 4.0 Smart Garment Factory Safety & Monitoring  
System

## **USER MANUAL**

## **Welcome to TaxyShield**

Welcome to TaxyShield, your intelligent assistant for maintaining a safe and efficient factory environment. This guide will help you navigate the system's features, whether you are managing the factory floor or operating machinery.

### **Getting Started**

When you open the application, you will be greeted by a welcome screen.

- Existing Users: Tap Login and enter your registered email and password to access your dashboard.
- New Users: Tap Register to create your account. You will need to provide your full name, email, a secure password, and select your designated role (Worker or Administrator).

### **Administrator Guide**

If you are logged in as an Administrator, your dashboard serves as a central command hub for monitoring factory health and safety.

- Factory Overview & Controls: Your dashboard displays summary cards giving you an instant glance at the factory's operational health. You can also monitor and toggle sewing floor device controls directly from this screen.
- Attendance & Security: Tap Attendance to monitor the live camera feed at the factory entrance and view secure, blockchain-verified attendance records. Tap "Refresh" to fetch the latest data.
- Safety Ledger: Access the Safety Ledger to review a highly secure, immutable list of past safety events and their verification statuses.
- Emergency Response: If an incident occurs, you will receive emergency alerts in your notifications. Open the alert to review the details, dispatch necessary assistance, and mark the situation as "Resolved" once it has been handled.
- PPE Verification: Open the PPE Check module to monitor the automated safety equipment checks happening at the factory entrances.

## **Factory Worker Guide**

If you are logged in as a Worker, your dashboard is streamlined to focus on your daily tasks, equipment maintenance, and personal safety.

- **Daily Tasks & Profile:** Check your dashboard for today's assignments. You can tap the Profile icon in the top navigation bar at any time to review your personal details or securely log out.
- **Health Monitoring:** Tap the Health Widget to view interactive graphs of your heart rate and body temperature over time. Pay close attention to this screen - if any limits are exceeded, the system will display colour-coded warnings advising you to take action.
- **Augmented Reality (AR) Maintenance:** To perform a maintenance check on your equipment, tap Scan Machinery (AR). Point your device's camera at the machinery and follow the holographic text and arrows that appear on your screen for guided assistance.
- **Emergency Reporting (SOS):** In the event of an accident or hazard, tap the red SOS button on your dashboard. Select the type of emergency, type in any optional details, and tap Submit Alert to instantly notify the administration team.

## **Automated PPE Clearance**

When entering the factory floor, an automated kiosk handles your safety clearance.

1. Stand clearly in front of the entrance camera.
2. The system will automatically scan you for required Personal Protective Equipment (PPE), such as masks and gloves.
3. If your equipment is worn correctly, you will be granted clearance. If anything is missing, the system will display a warning telling you exactly which equipment you need to put on.

## Daily Log Sheet

Table 5.2: Daily Log Sheet

| Week/Date                                                                    | Activities Completed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Lecturer's Comments                                                                                                                                                                      | Lecturer's Signature |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Proposal Presentation<br>(27 <sup>th</sup> of May - 2026)                    | Finalized the project idea, identified its key features, and selected four emerging technologies for implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Add more features: stress, burnout detector, predictive maintenance - machine failure alerts                                                                                             |                      |
| 1st progress<br>(23 <sup>rd</sup> /24 <sup>th</sup> of June - 2026)          | <p>We completed the hardware component by integrating IoT sensors (gas, light, current, motion, and temperature) connected to an ESP32 microcontroller to monitor real-time factory conditions and detect fire risk and machine overcurrent.</p> <p>We configured the system to automatically control devices such as lights, air conditioning, exhaust fans, water pumps, and the main power supply in response to hazardous situations like fire, gas leaks, and electrical overloads.</p> <p>We also integrated Artificial Intelligence to predict fire risks, detect gas anomalies, and identify overcurrent conditions.</p>                                                                                                                                          | Hardware 50% complete; the next step is to integrate and verify the machine and human change prediction features using the mobile app. Do more research related to factory environments" |                      |
| 2nd Progress<br>(16 <sup>th</sup> of July - 2026)                            | <p>Started development of the Kotlin mobile application and successfully implemented user authentication.</p> <p>Built separate Admin and Worker side interfaces.</p> <p>Integrated Firebase for real-time data storage; hardware sensor readings and risk predictions are displayed on the admin page via Firebase.</p> <p>Structured the Worker-side layout and profile page.</p> <p>Designed a consistent UI/UX theme across the application.</p>                                                                                                                                                                                                                                                                                                                      | Hardware is 50% complete, and the AR feature for machine details has been implemented. Camera-based human detection has not yet started.                                                 |                      |
| Final Progress<br>[Viva / Presentation]<br>(30 <sup>th</sup> of July - 2026) | <p>Completed the hardware integration by adding the ESP32-CAM module to capture worker QR code details, with entries verified and recorded on the admin side through a blockchain-based process, including a live camera feed for real-time monitoring.</p> <p>Completed AI training with image pre-processing for PPE (mask) detection and worker health detection.</p> <p>Implemented machine maintenance detection using AR technology to display machine details and user manuals through the app.</p> <p>Integrated an interactive, responsive, and polished UI/UX, including a landing page.</p> <p>Finalized the application name.</p> <p>Completed the mobile application and hardware components.</p> <p>Secured a client to partner with us on the project.</p> | Overall, the project is good - the hardware, AR machine details, and stress detection all work well                                                                                      |                      |

